



Impact Analysis of Cyber Attacks on Power System Operation: Cascading Failures and Blackout

Mohamed Radwan, Dr. Alex Stefanov

EE4410 - Cyber Security of Power Grids

Faculty of Electrical Engineering, Mathematics and Computer Science, **TU Delft**

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Outline

1. Power system dynamics: modelling and simulation

- Dynamics and stability
- Protection modelling and coordination

2. Cascading failures and blackouts

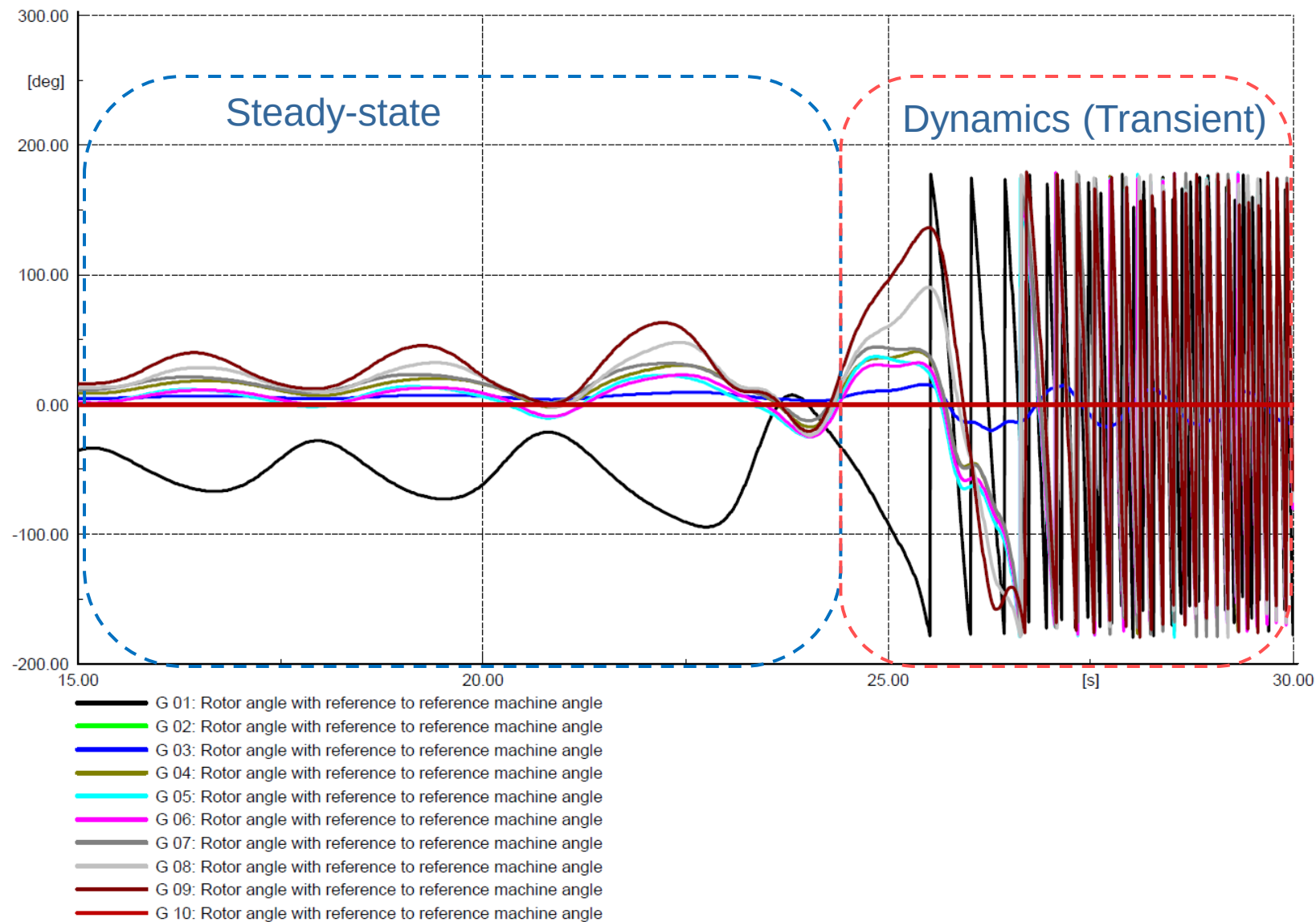
- Historical blackouts
- Cascading failure mechanism
- Cyber attack triggers

3. Power grid cyber security and impact analysis

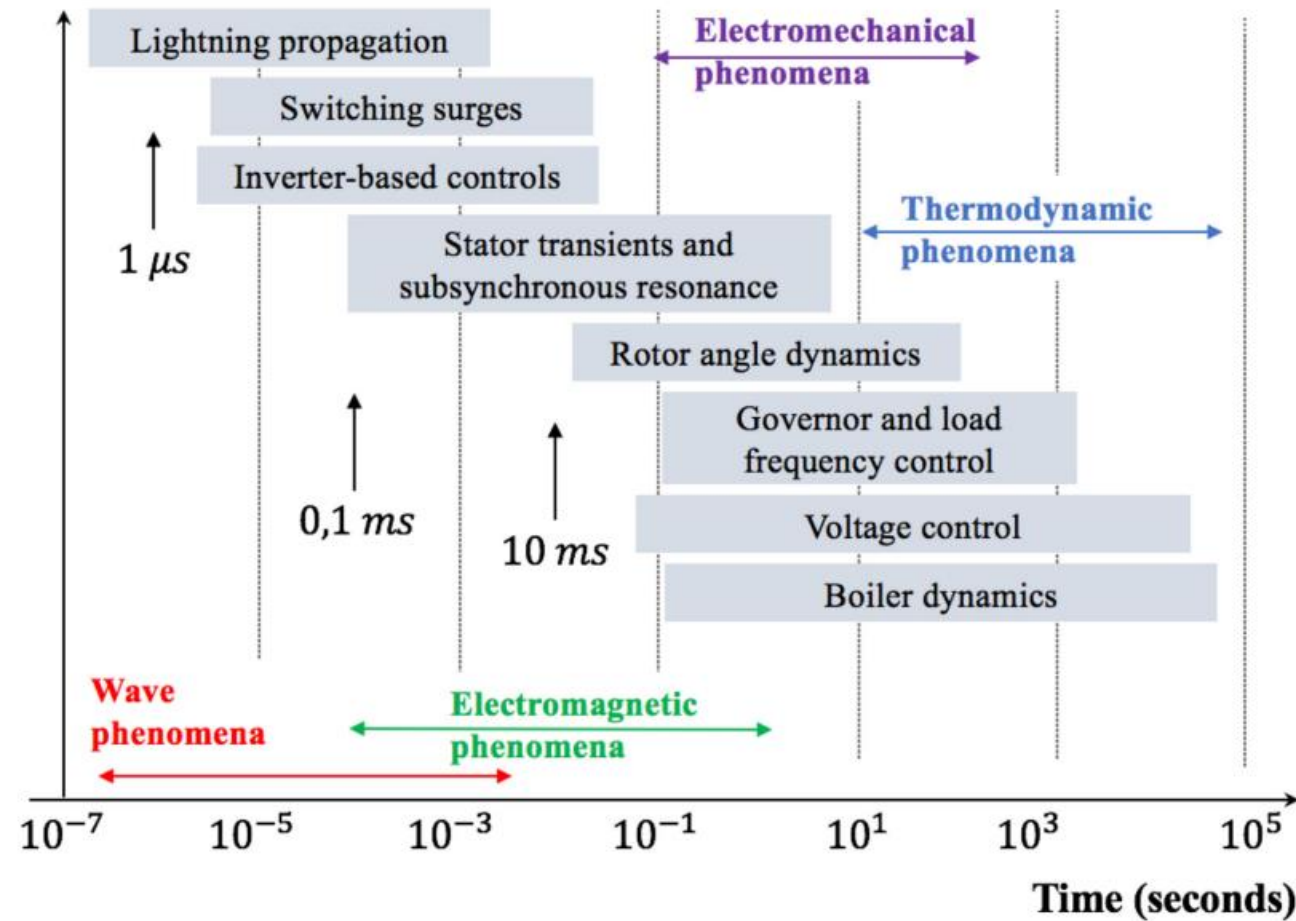


1. Power System Dynamics: Modelling and Simulation

Power System Dynamics



Power System Dynamics: Timescales

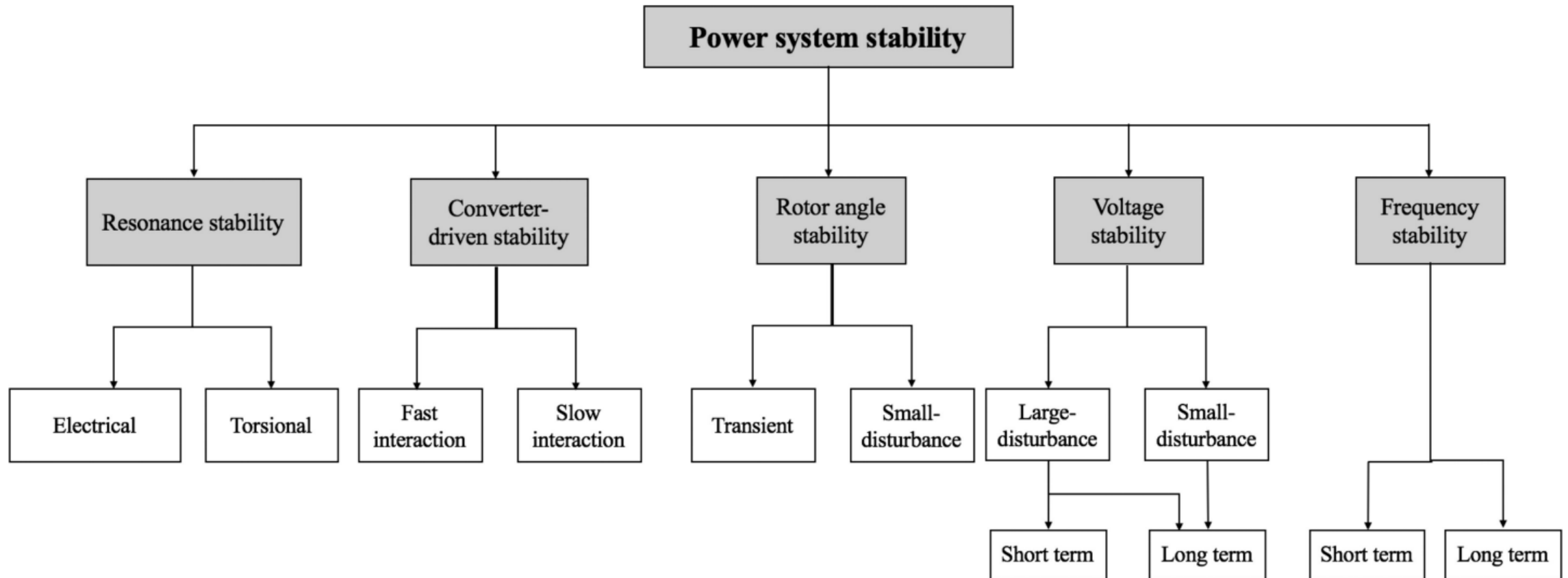


What is Power System Stability?

- Maintain **state of equilibrium** under nominal operating conditions
- Return to **state of equilibrium** post **disturbance**
- Power grid is **nonlinear and non-stationary** system!
- Disturbances:
 - Small disturbances, e.g., changes in loading
 - Large disturbances, e.g., faults, loss of equipment



Classification of Power System Stability



Numerical Simulations to Analyse Power System Dynamics

- Power system dynamical behaviour characterised by two main properties:
 - nonlinearity – sinusoidal or nonlinear functions
 - nonstationarity – varying w.r.t *time*, i.e., *function* (*t*)
- Described by a general set of Differential Algebraic Equations (DAEs)

$$\dot{x} = f(x(t), y(t), z(t))$$

$$g(t, (x(t), y(t), z(t))) = 0$$

$$h(t, x(t), y(t), z(t)) < 0$$

- Simulation solvers to integrate and solve DAEs
- Various numerical methods to solve; trapezoidal rule, Euler's method, R-K method, etc.

Power System Dynamics: Protection Modelling

- Immune system of power grid
- Capture system response to disturbances/events
- Simplified representation:

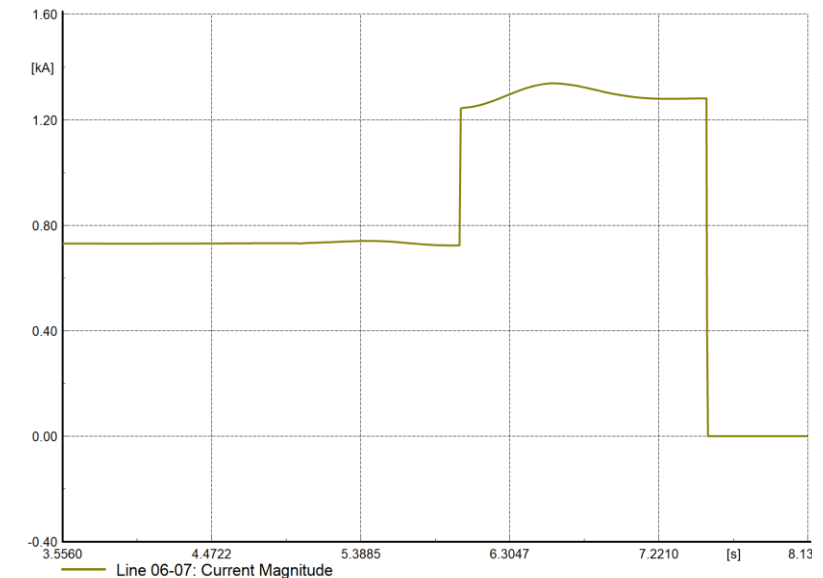
$$K_s = \begin{cases} 0, & 0 < I_s \leq I_{\text{pickup}} & t < t_{\text{lim}} \\ 1, & I_{\text{pickup}} < I_s < I_{\text{lim}} & t > t_{\text{lim}} \end{cases}$$

Protection Scheme	Applied on
Over/Under Frequency	Generators
Over/Under Voltage	Generators
Rate of Change of Frequency (ROCOF)	Generators
Overflux	Generators
Pole-slip (out of step)	Generators
Underfrequency Load Shedding	Loads
Undervoltage Load Shedding	Loads
Distance	Lines
Overload	Lines

Protection Models: Distance Protection

- Primary protection of lines against fault and over-currents
- Zone reach for each distance relay defined based on the protection coordination assistant of PowerFactory
- Equations for each zone and time setting, calculated as follows:

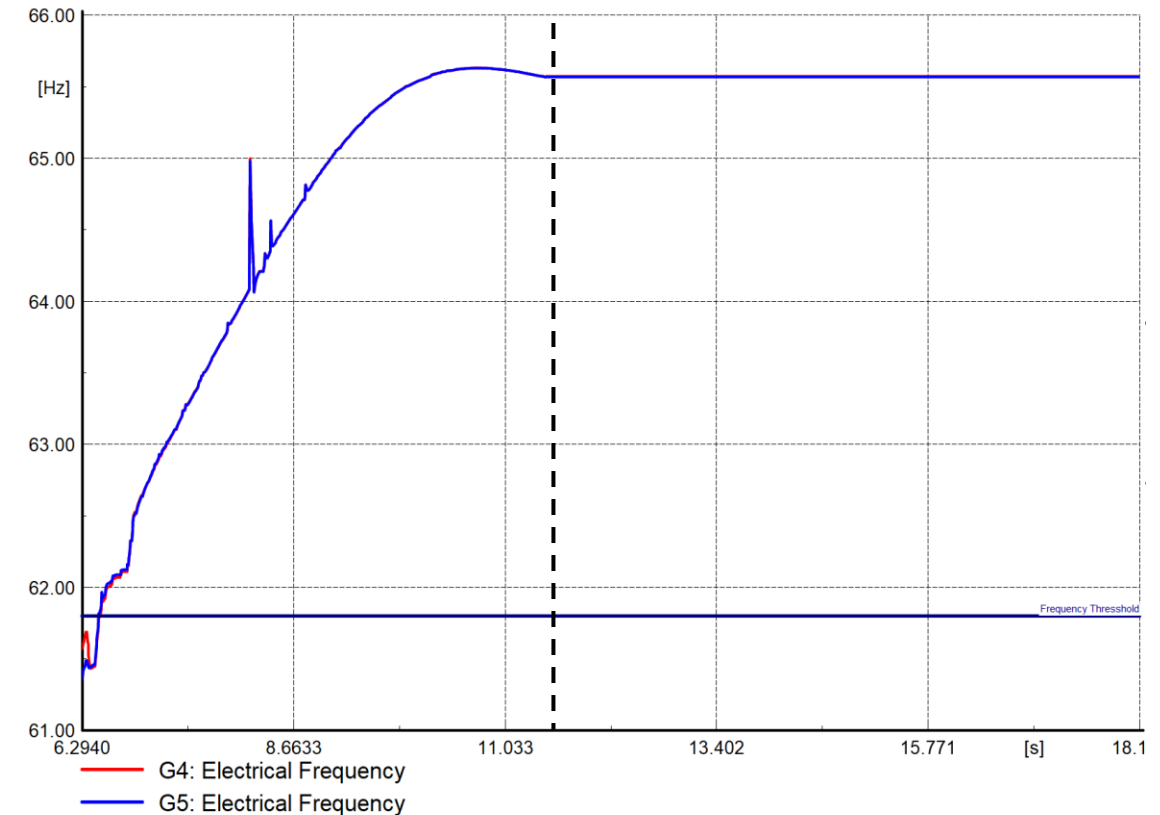
Zone	Zonal Impedance (Ohms)	Time Setting (s)
1	$Z_1 = 0.85 * Z_{minA-B}$	0
2	$Z_2 = Z_{maxA-B} + 0.6 * Z_{minB-C}$	0.6
3	$Z_3 = Z_{maxA-B} + Z_{minB-C} + 0.2 * Z_{minC-D}$	0.8



Protection Models: Over/Under Frequency Generator Protection

Over frequency Threshold (Hz)	Setting (s)
> 60.60	600
> 61.0	100
> 61.8	5

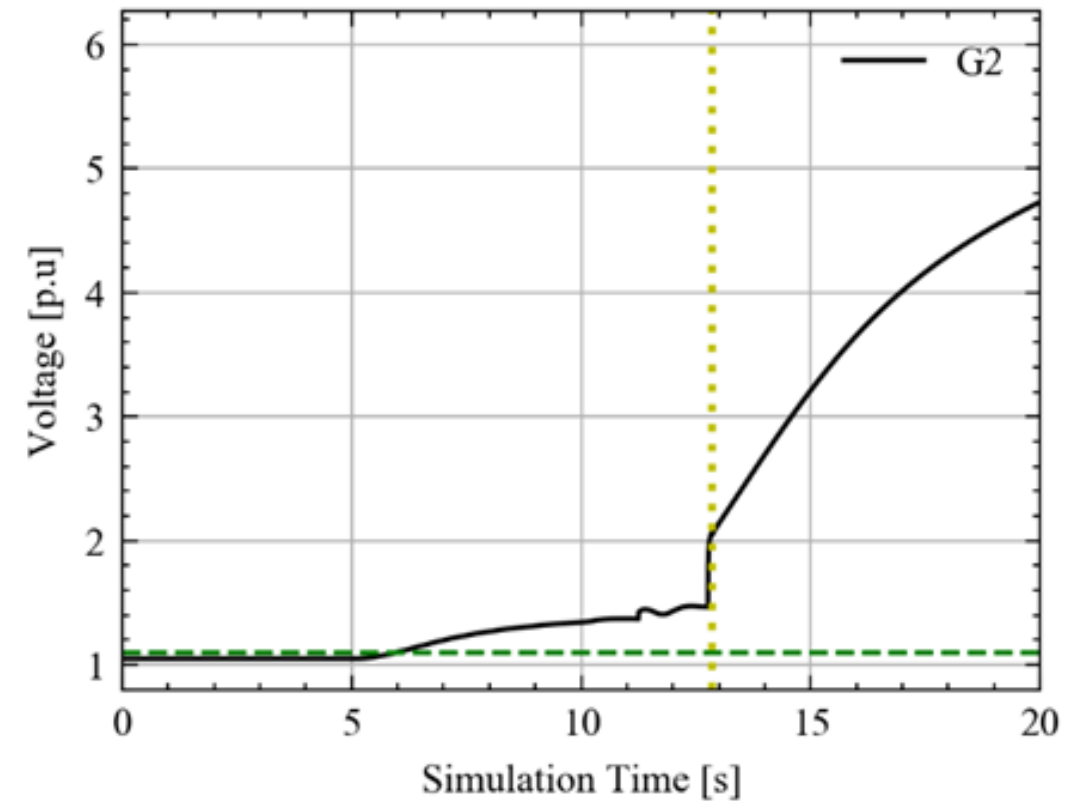
Under frequency Threshold (Hz)	Setting (s)
< 58.4	90
< 57.7	30
< 57.1	0.167 (10 cycles)



Protection Models: Over/Under Voltage Generator Protection

Over voltage Threshold (p.u.)	Setting (s)
> 1.1	15
> 1.5	0.083 (5 cycles)

Under voltage Threshold (p.u.)	Setting (s)
< 0.9	15
< 0.8	2



Protection Models: ROCOF Generator Protection

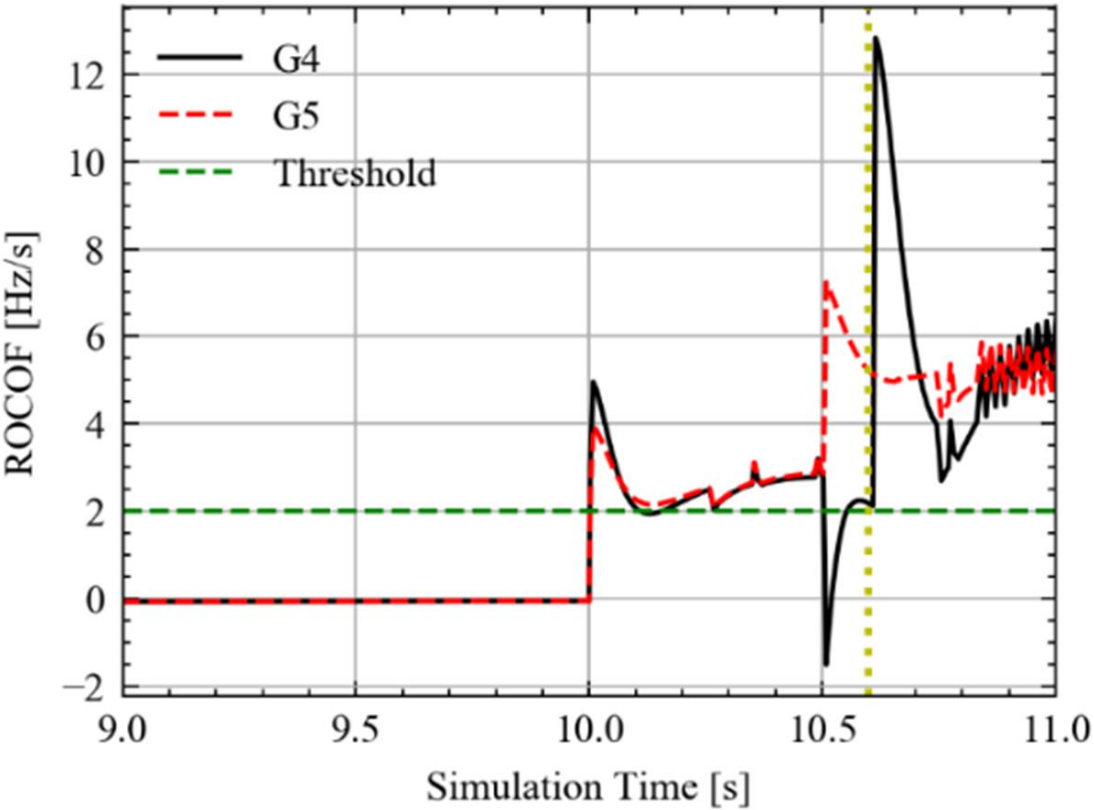
ROCOF settings per different European grid codes

Country	RoCoF limit [Hz/s]	Time to disconnect [s]
Netherlands	2.0	0.5
Italy	2.5	0.1
Finland	2.0	0.5
Greece	0.5	Not mentioned
Ireland	1	0.5
United Kingdom	1	0.5

Protection Models: ROCOF Generator Protection

ROCOF Threshold (Hz/s)	Setting (s)
> 2.0	0.5
< -2.0	0.5

Sampling time = every 60 [ms]



Protection Models: Underfrequency and Undervoltage load shedding

Underfrequency Threshold (Hz)	Setting (s)	Load Shed (%)
< 59.1	5	5.3
< 58.9	2.5	5.9
< 58.7	1	6.5
< 58.5	0.1	6.7

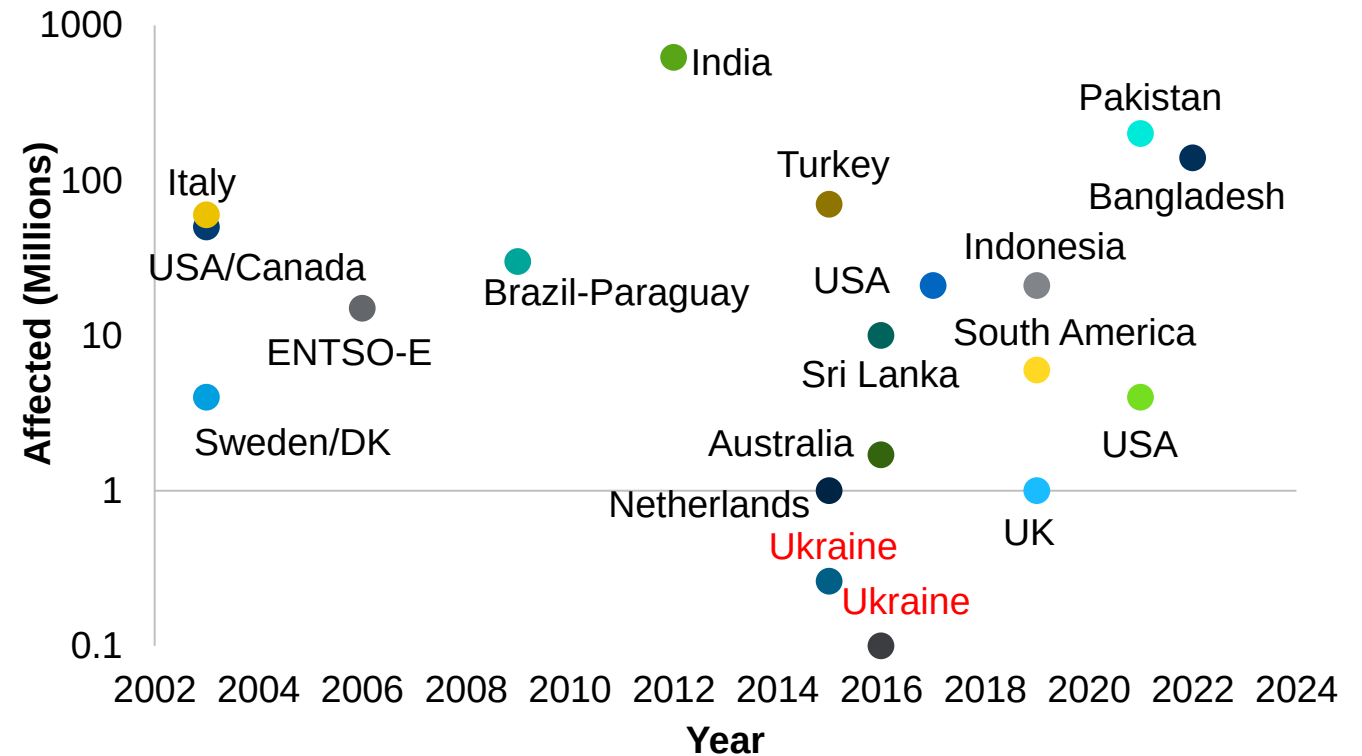
Undervoltage Threshold (p.u)	Setting (s)	Load Shed (%)
< 0.8	30	5



2. Cascading Failures and Blackouts

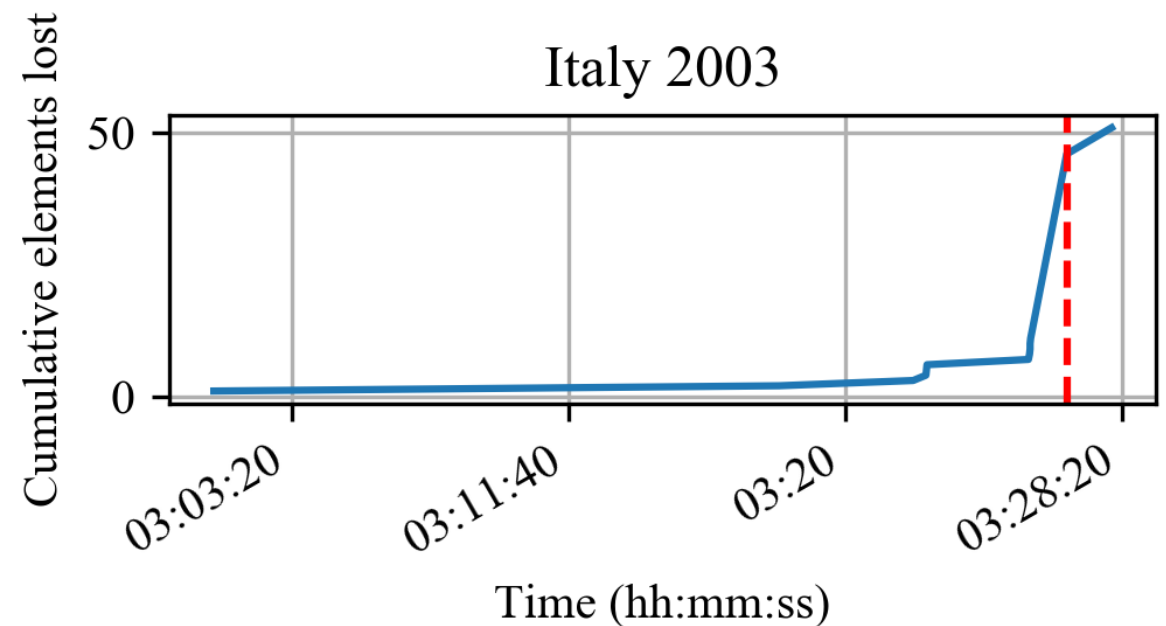
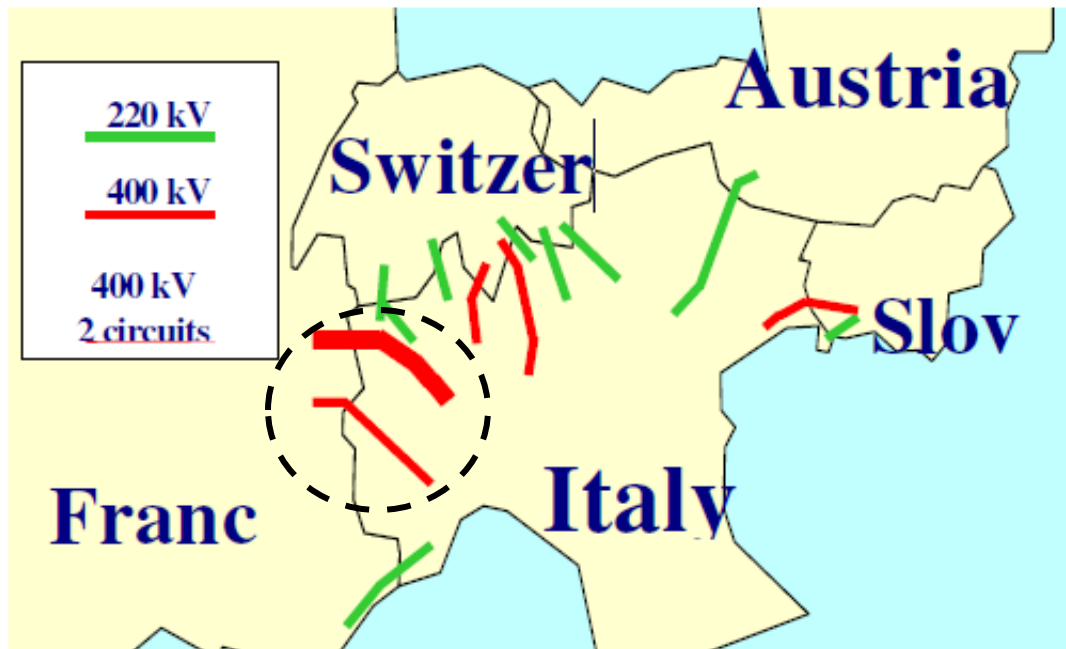
Global Power System Outages

- Power outages more frequent
 - Increasing energy demand
 - energy transition
- Common link: cascading failures
- Combination of multiple, mutually exclusive, low probability events
- Perfect storm conditions



Historical Blackouts: Italy 2003

- Initiated by 380 kV tie line tripping in Switzerland due to a flashover, close to the Italian border
- Neighbouring lines overloaded
- Subsequent line tripping, cascading failures and blackout
- 6.4 GW unserved, 60 million affected

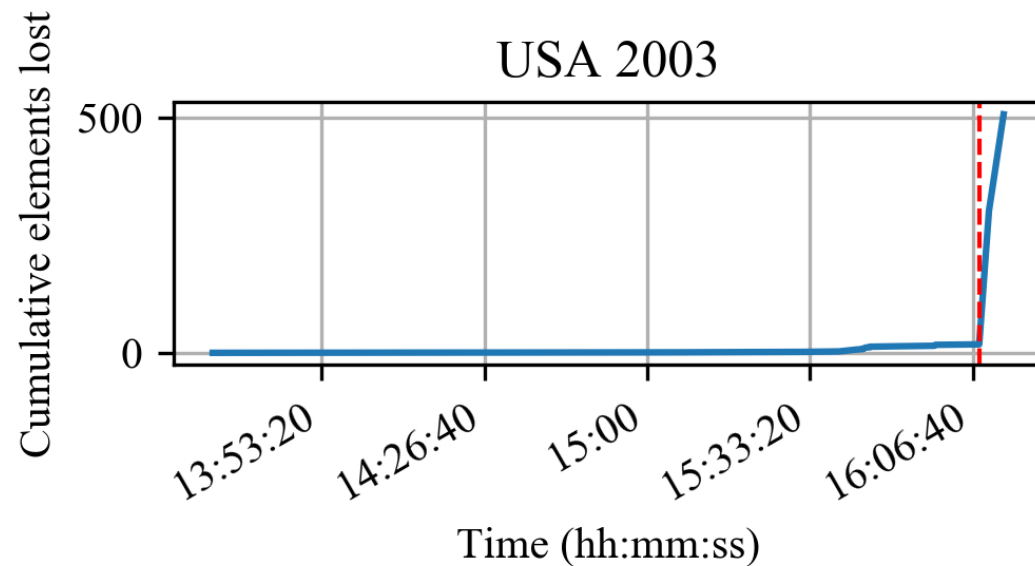


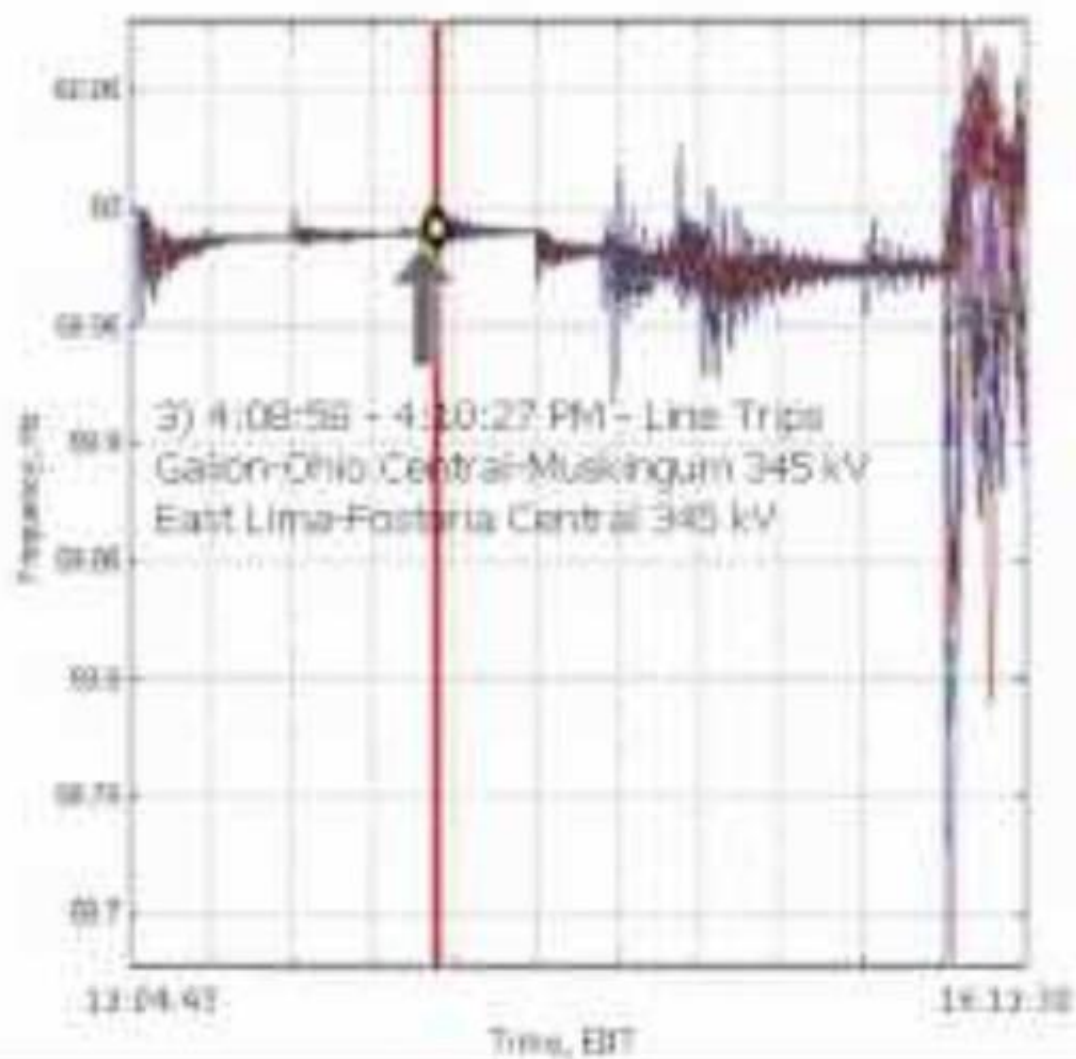


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Historical Blackouts: USA-Canada 2003

- Root cause: **software bug** in utility – lack of situational awareness
- Multiple transmission lines tripped due to contact with vegetation – set off **cascading** chain
- ~400 transmission lines and 531 generating units at 261 power plants tripped
- **63 GW** unserved, **50 million** affected

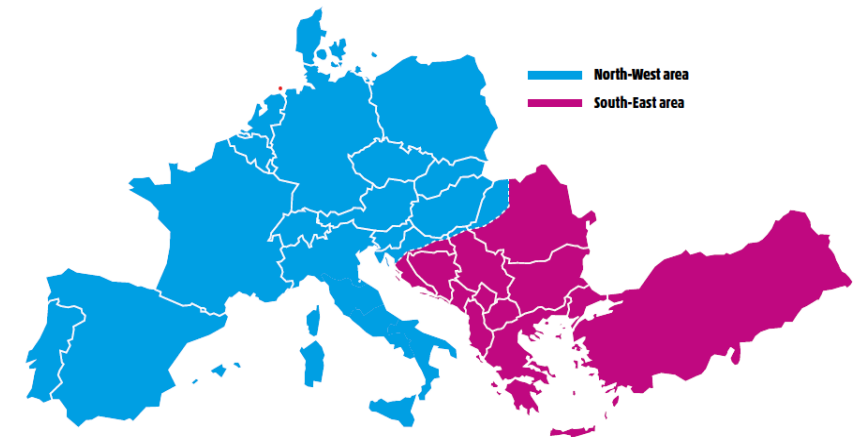
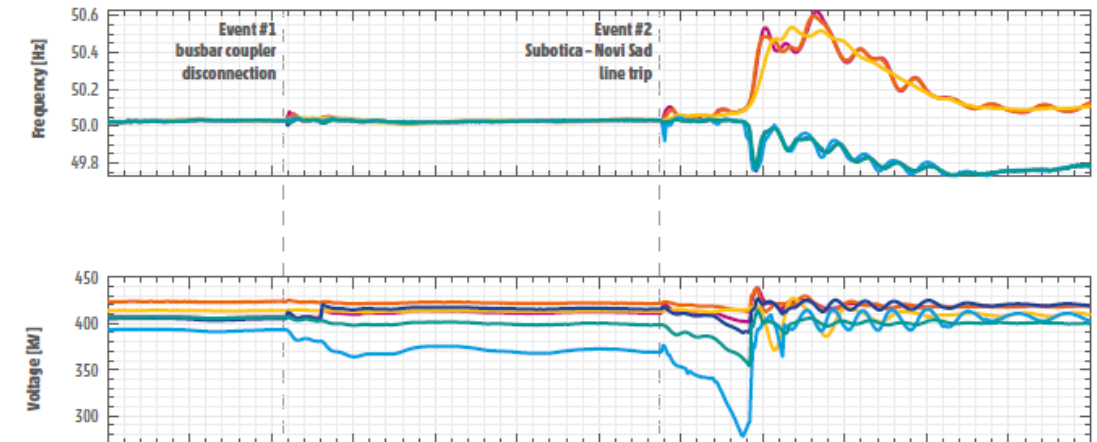




ENTSO-E System Split: January 2021

- Initiated by a busbar coupler trip in Croatia
- Cascading tripping of transmission lines
- *Point of no return* reached in ~ 20 seconds!

No	TSO	delta/s	trip time	substation 1	substation 2	voltage/kV	Comments
1a	HOPS	0	14:04:25.9	Ernestinovo		400	busbar coupler overload protection
1b	HOPS	2.6	14:04:28.0	Ernestinovo		400/110	Overload protection of both 400/110 kV transformers
2	EMS	23	14:04:48.9	Subotica	Novi Sad	400	overload protection 20 s 2 nd zone
3	TRANS	26	14:04:51.9	Paraşeni	Târgu Jiu Nord	220	distance prot. starting zone 2.4 s
4a	TRANS	27.9	14:04:53.8	Reşiţa	Timişoara	220	dist. prot. 0.4 s
4b	TRANS	27.9	14:04:53.8	Reşiţa	Timişoara	220	dist. prot. 0.4 s, breaker L1 failure
5	NOS BiH	28.2	14:04:54.1	Prijedor	Međurić	220	dist. prot. out-of-step protection
6	NOS BiH	28.2	14:04:54.1	Prijedor	Sisak	220	dist. prot. out-of-step protection
7	HOPS	28.3	14:04:54.2	Melina	Velebit	400	dist. prot. zone 3
8	TRANS	28.3	14:04:54.2	Mintia	Sibiu	400	distance prot. power swing cond.
9	HOPS	28.5	14:04:54.4	Brinje	Padene	220	dist. prot. zone 1
10	TRANS	28.6	14:04:54.5	Gădălin	Iernut	400	distance prot. power swing cond. zone 2 0.4 s
11	TRANS	28.7	14:04:54.6	Sibiu Sud	Iernut	400	dist. prot. zone 3 reverse 0.6 s
12	TRANS	28.7	14:04:54.6	Autotransf 400/220	Roşiori	400/220	dist. prot. power swing cond.
13	TRANS	42.6	14:05:08.5	Iernut	Câmpia Turzii	220	dist. prot. zone 2 power swing conditions
14	TRANS	42.7	14:05:08.6	Fântânele	Ungheni	220	dist. prot. zone 2 power swing conditions

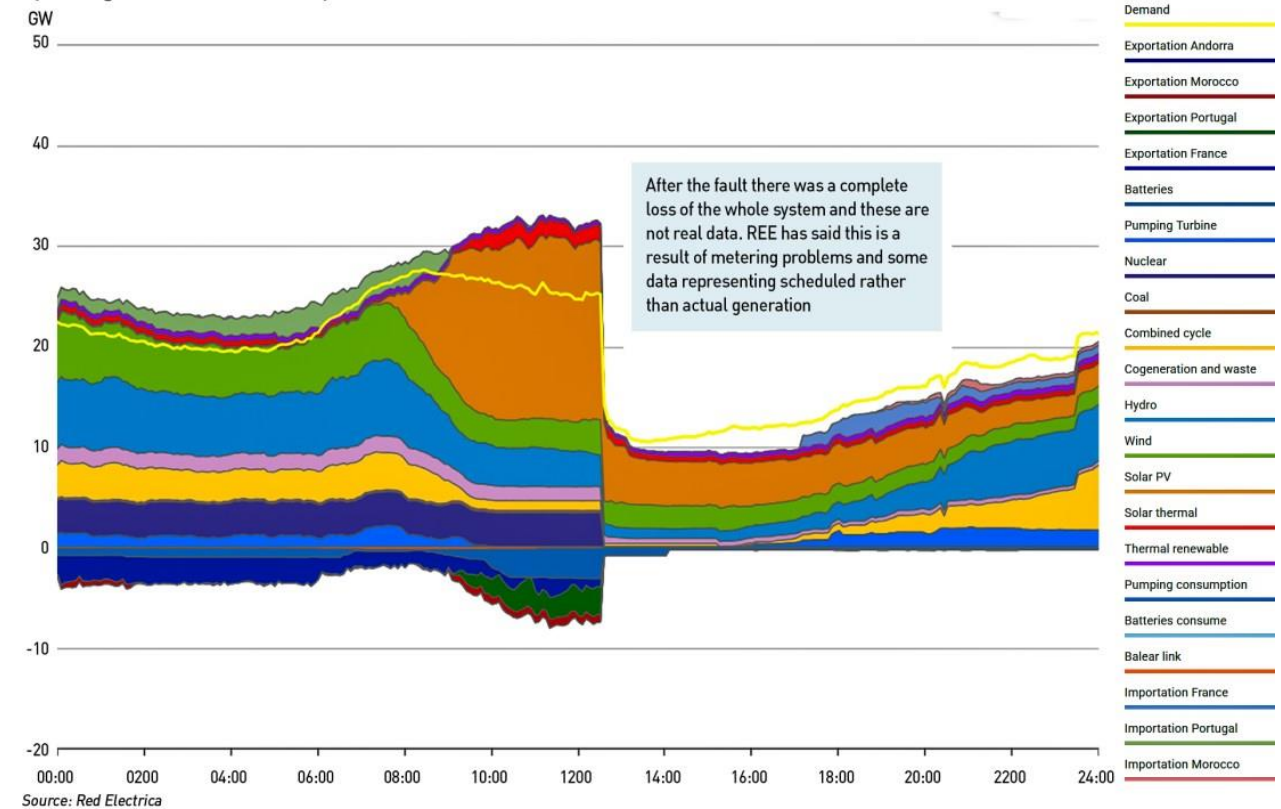


Iberian Peninsula Blackout: 28 April 2025

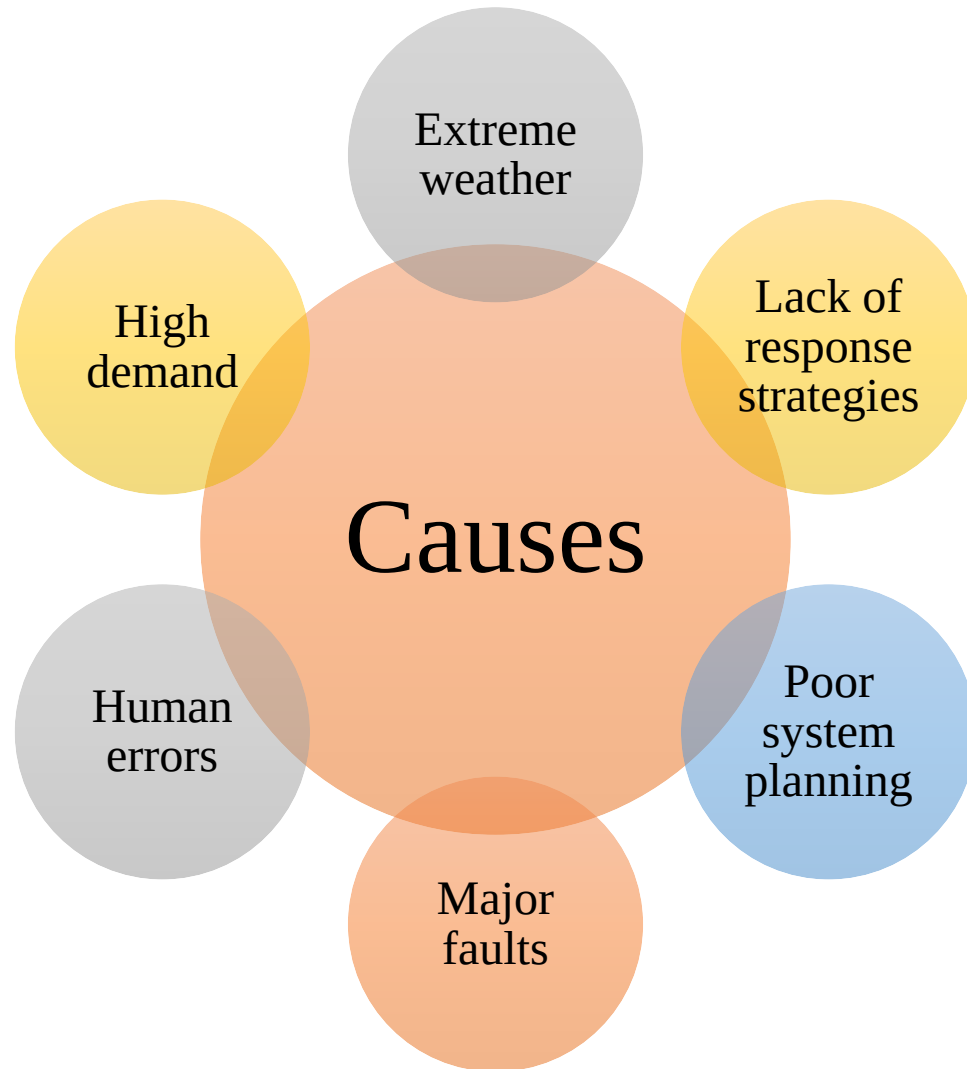
- Low System inertia (54.86% solar generation)
- Power and frequency swings
- **Cascading tripping** of generators (2200 MW lost)
- Tie Lines between France and Spain disconnected by protection
- Sudden loss of 15 GW within few seconds



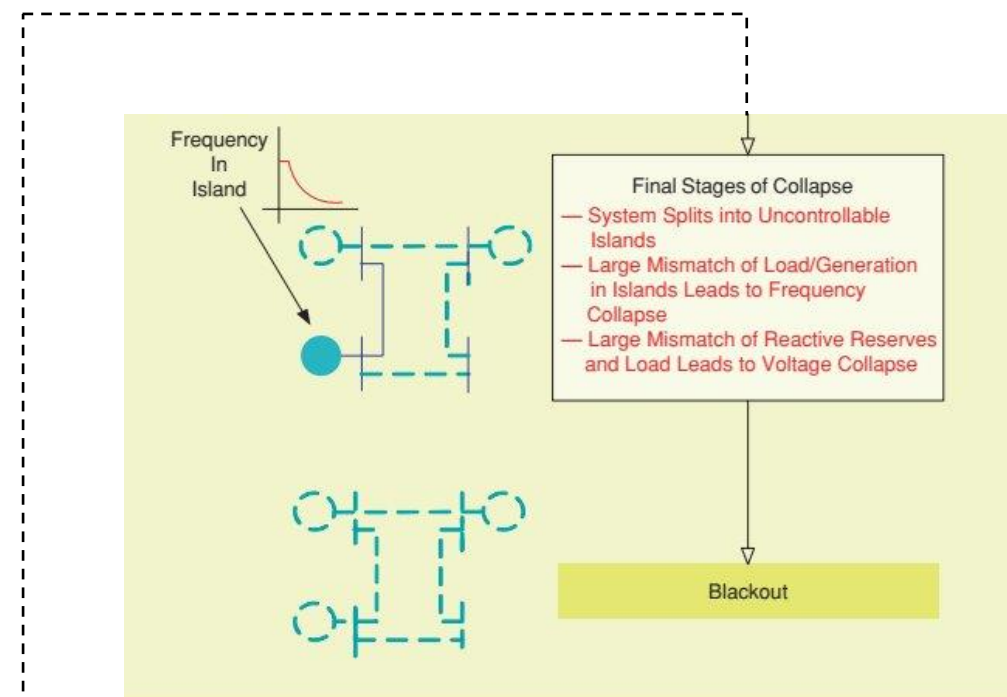
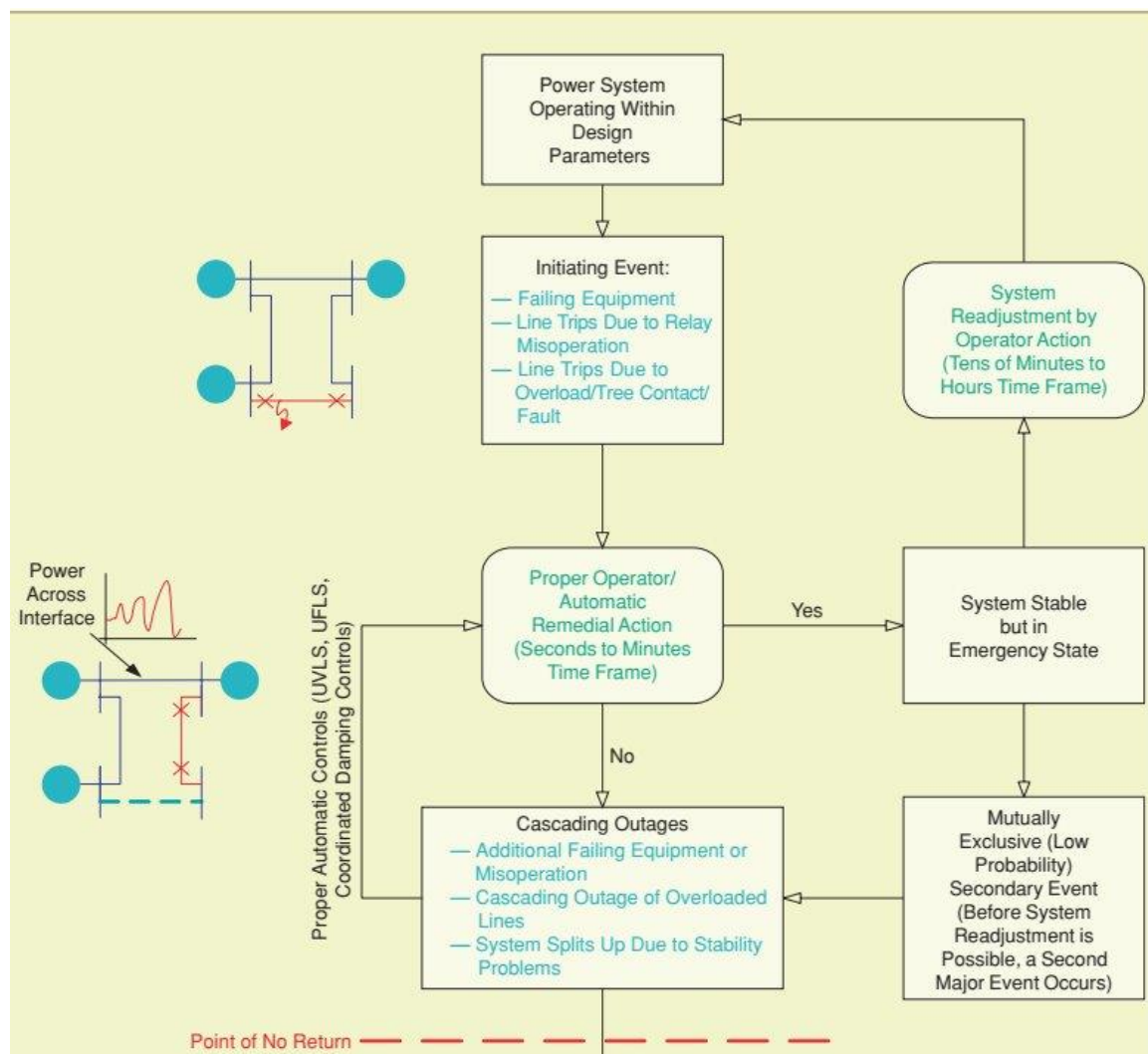
Spanish generation mix on 28 April 2025



What Causes a Blackout?

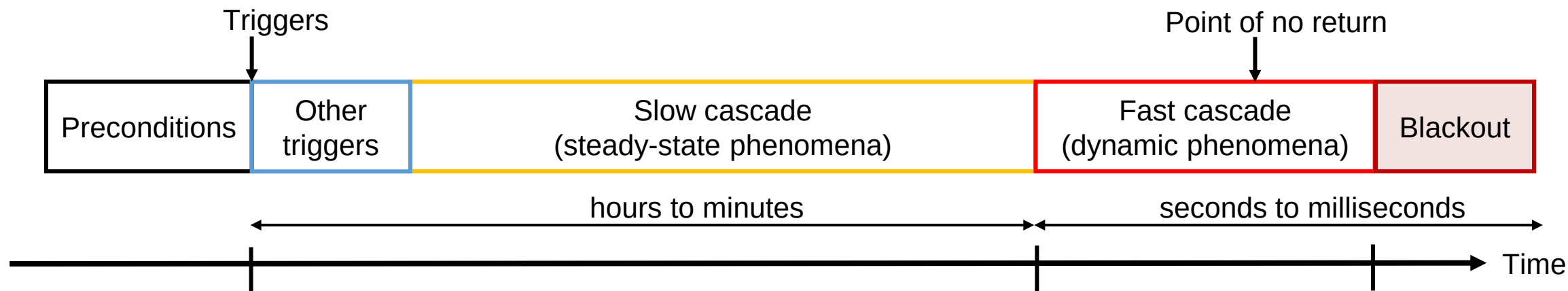


Anatomy of a Blackout



Anatomy of a Blackout: Brief Timeline

- Two distinct phases for most observed blackouts: *slow* vs *fast* phase



Event	Slow phase	Fast phase
USA/Canada 2003	1 h	5 min
Italy, 2003	24 min	2 min
Greece, 2004	12 min	2 min
Turkey, 2015	-	3 s

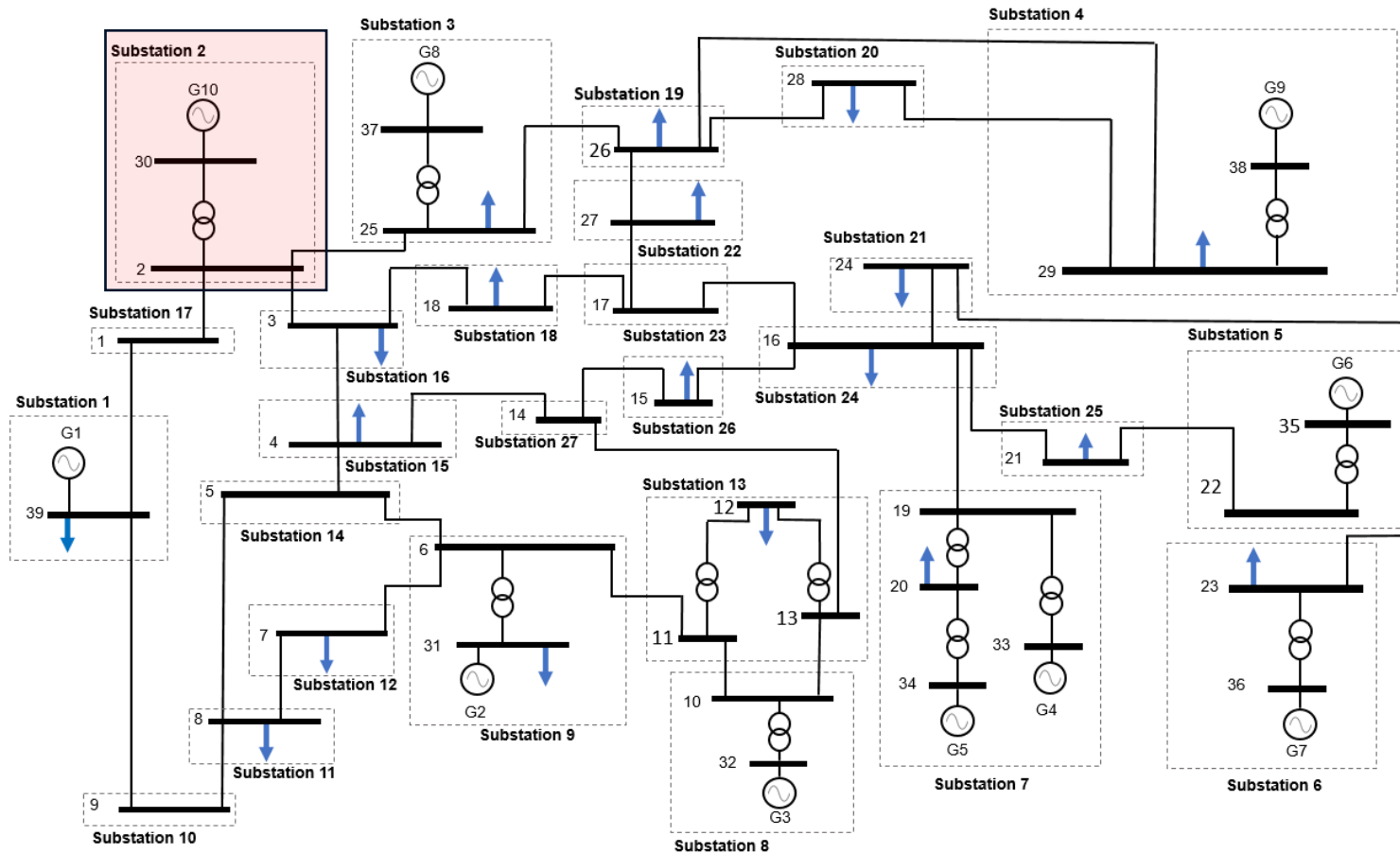
Critical Cyber-Physical Factors

Categories	Factors
Loss of transmission lines	Zone 3 distance protection
	Line overloading
	Protection malfunction
Voltage instability	Reactive power compensation
	Voltage regulation
	Line overloading
	Generator excitation controls
Transient instability (Loss of synchronism)	Fault clearing times
	Loss of generation
	Generator excitation controls
Frequency instability	Islanding
	Power mismatch
	Load shedding



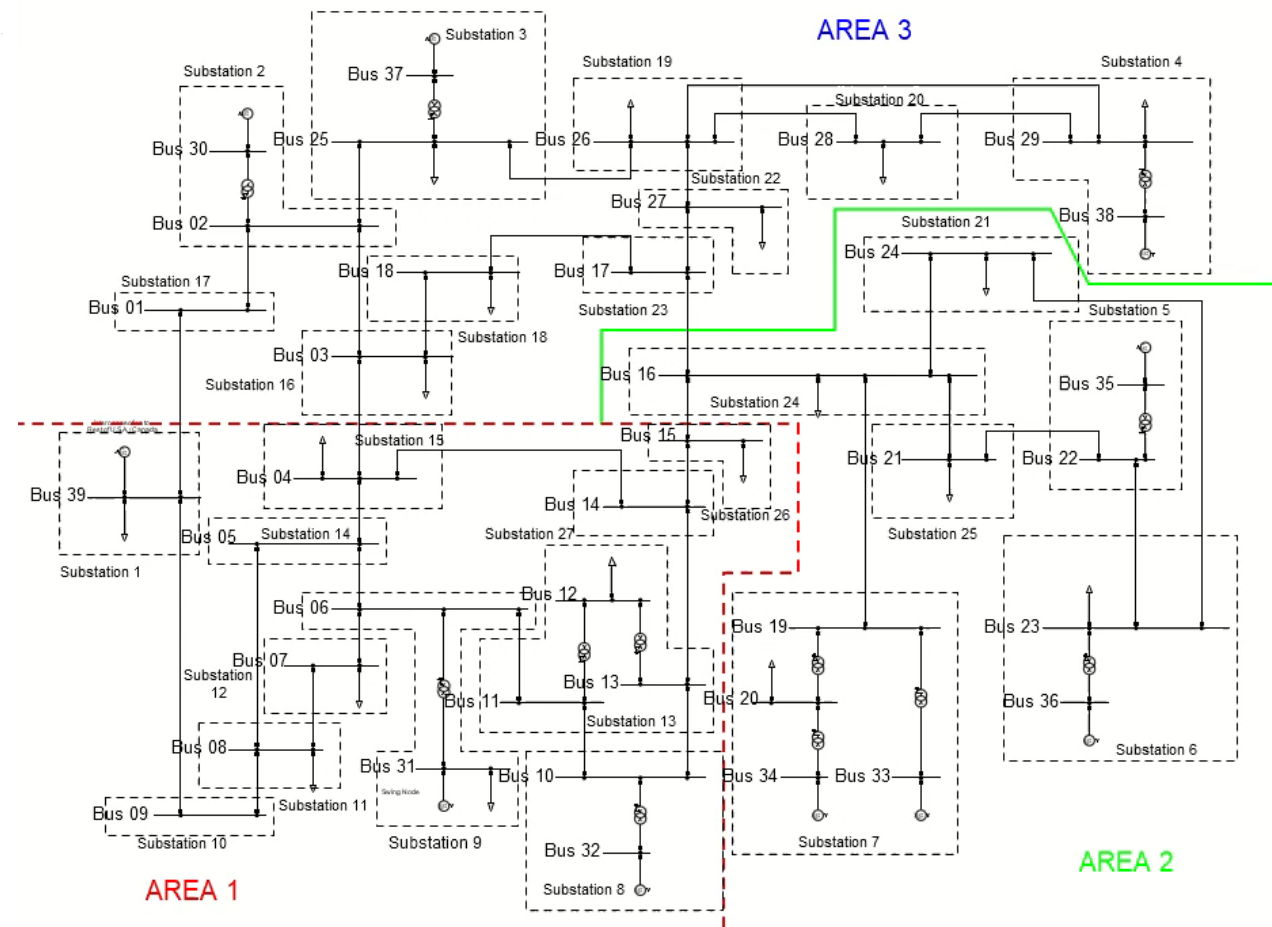
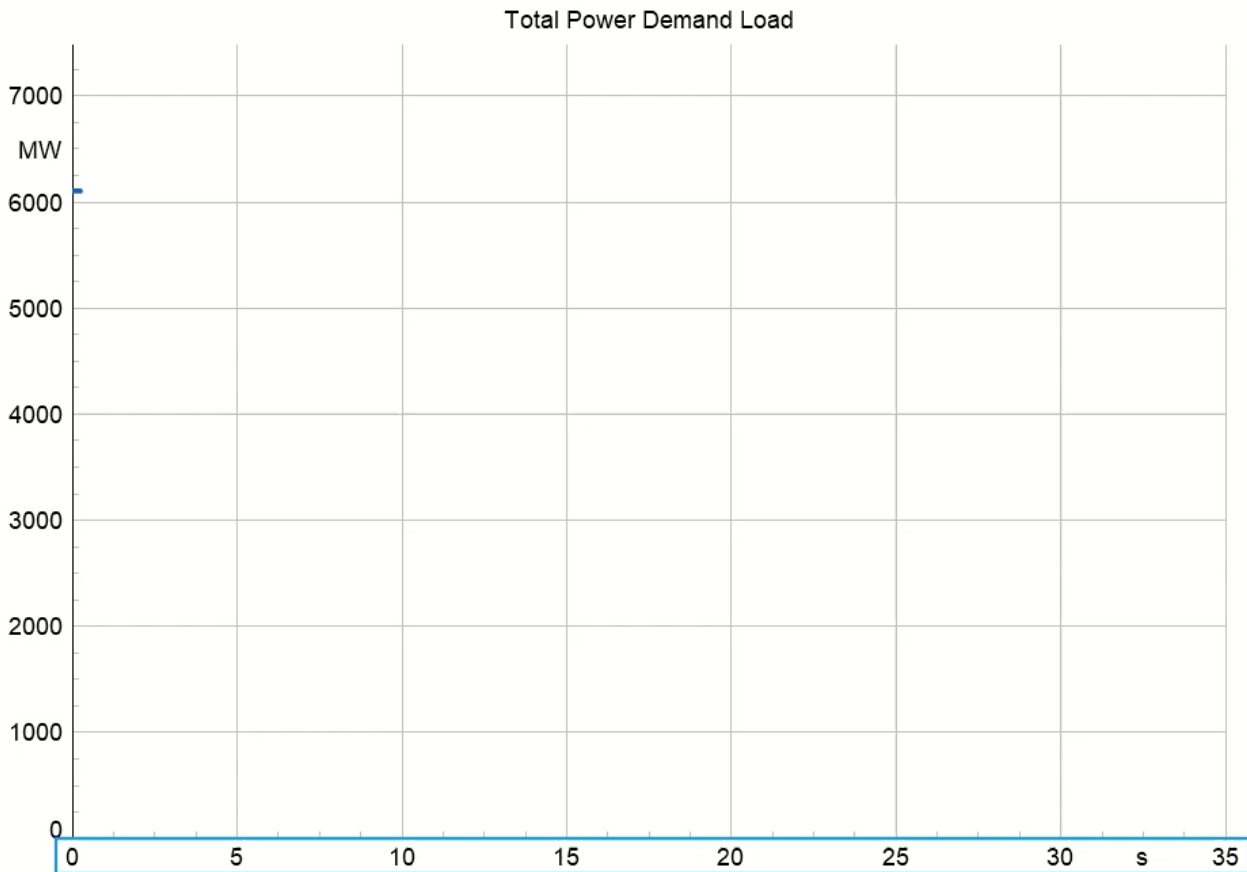
3. Impact of Cyber Attacks on Power System Stability: Cascading Failures and Blackout

Cascading Failure Analysis: Attack on Substation (S2)



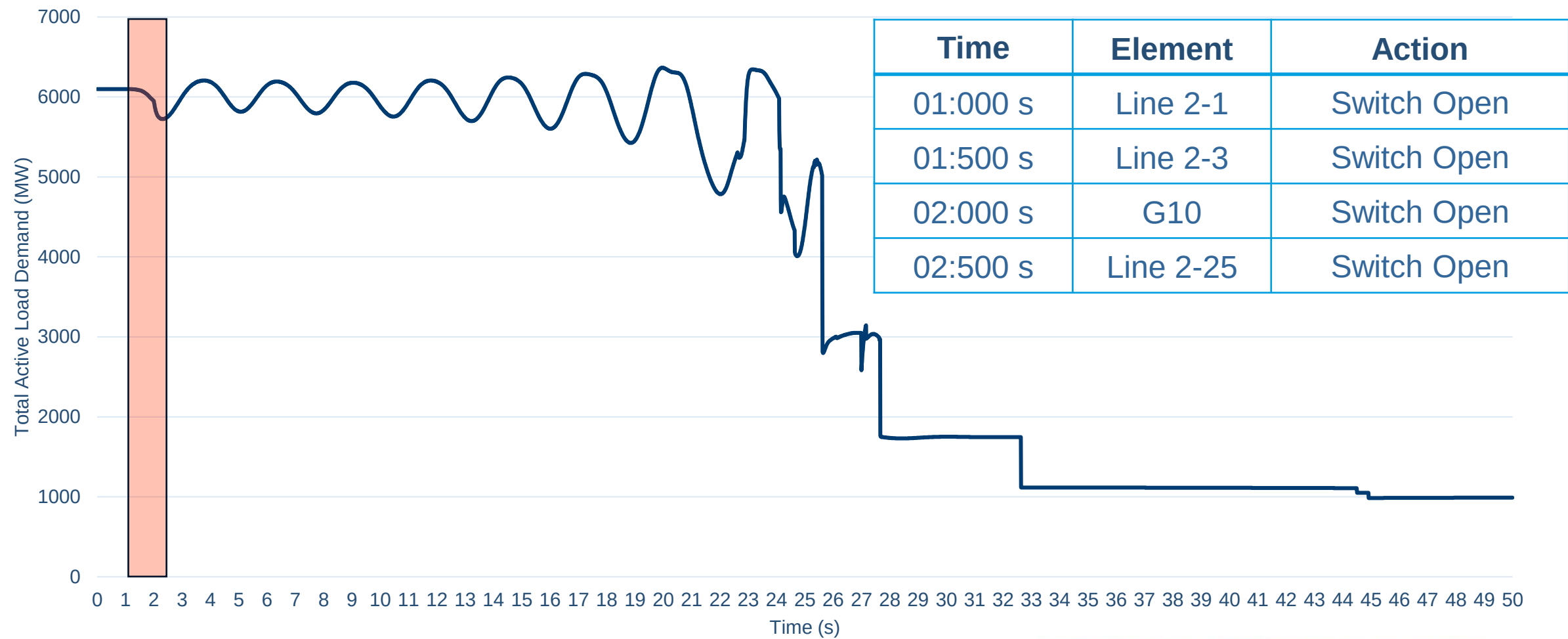
- Circuit breakers of lines 01-02, 02-03, 02-25, and Generator 10 in substation 2 are maliciously opened
- The attacks on substation 2 are launched at 1 to 2.5 seconds (simulation time)
- Simulation run time is 50 seconds (integration time-step = 10 ms)

Cascading Failure Analysis: Attack on Substation (S2)



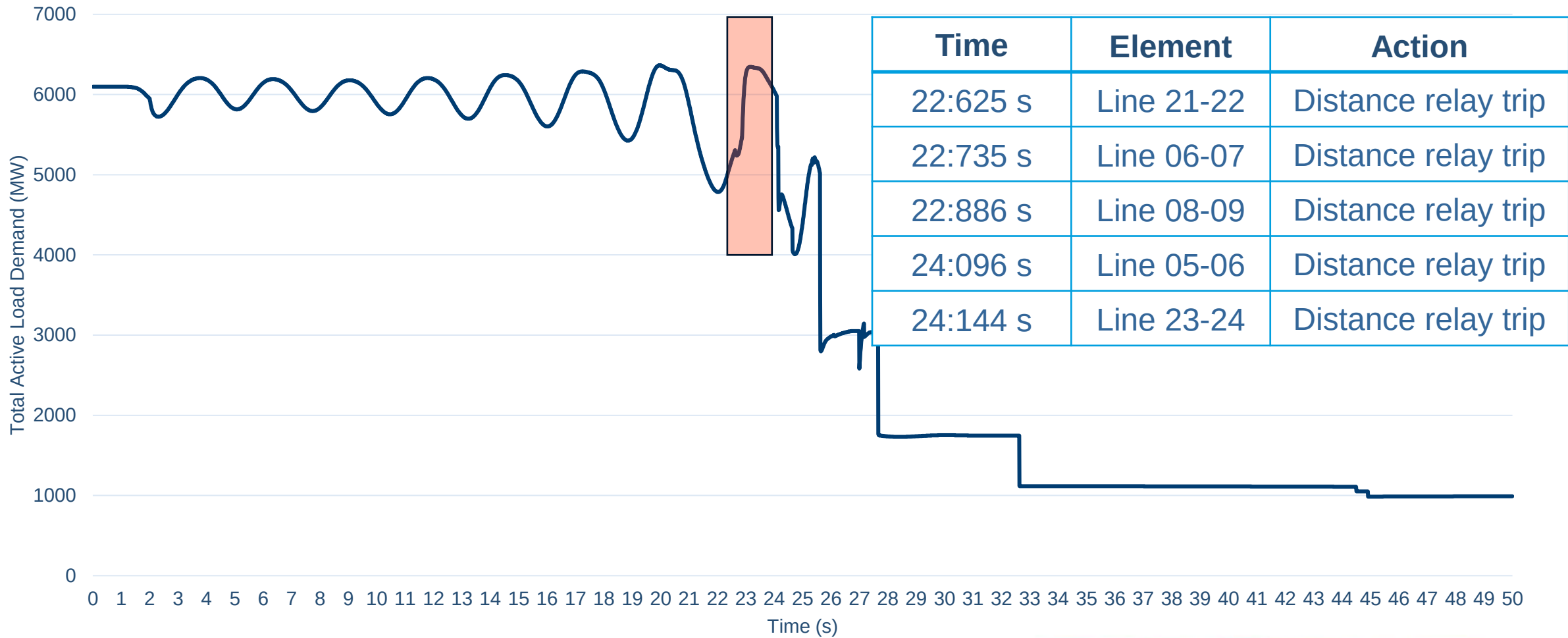
Cascading Failure Analysis: Attack on Substation (S2)

IEEE 39 Power Grid Loads



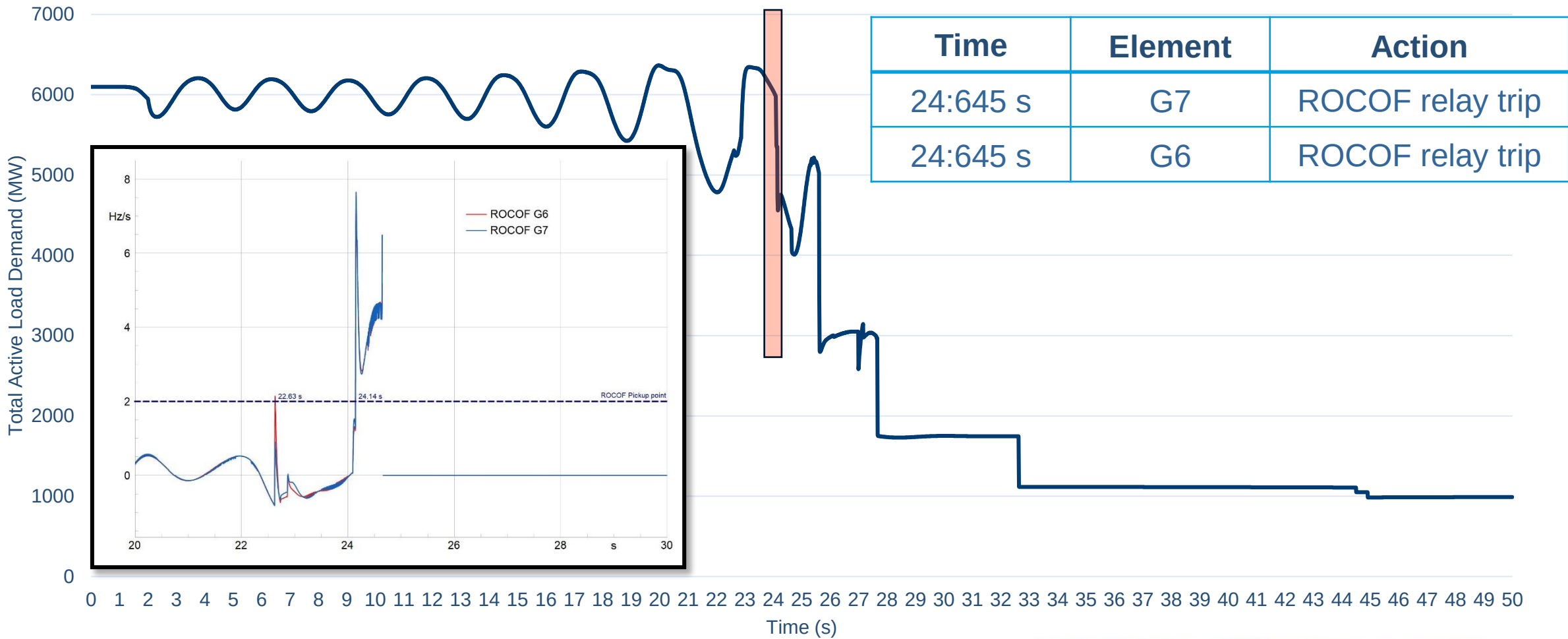
Cascading Failure Analysis: Attack on Substation (S2)

IEEE 39 Power Grid Loads



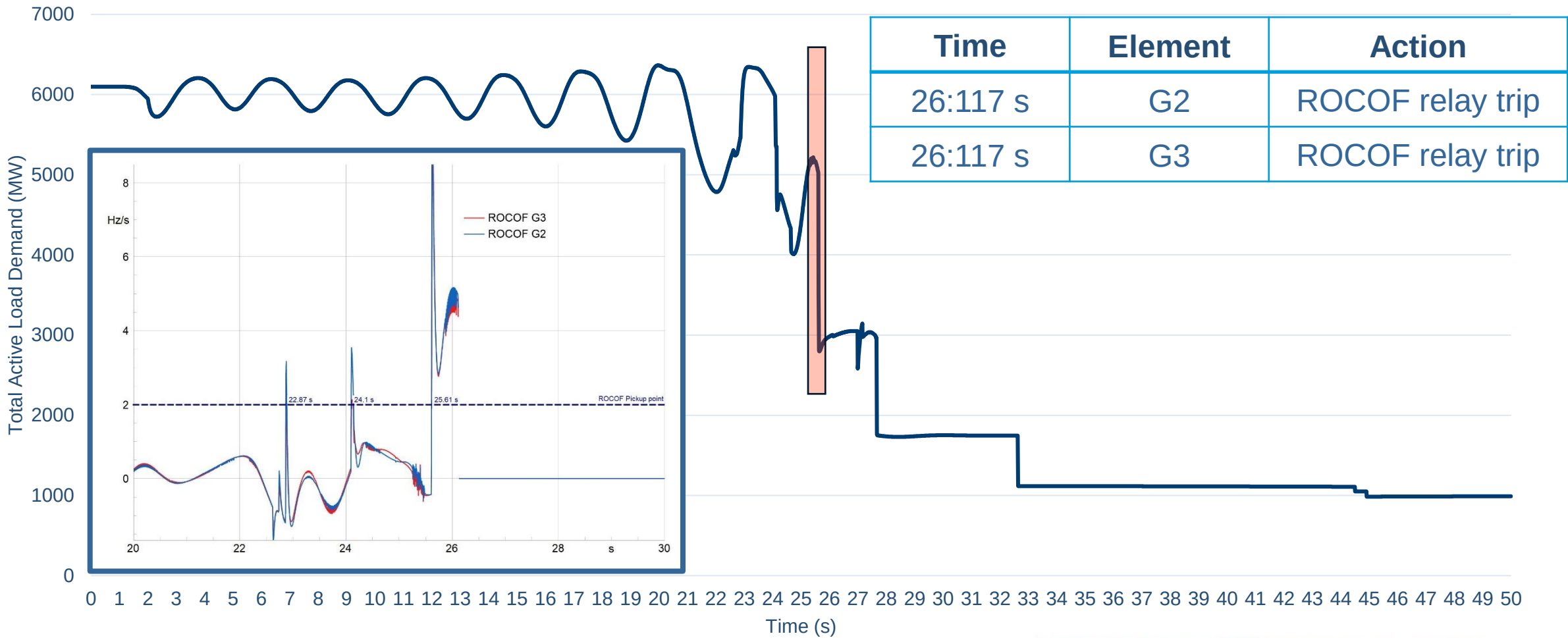
Cascading Failure Analysis: Attack on Substation (S2)

IEEE 39 Power Grid Loads



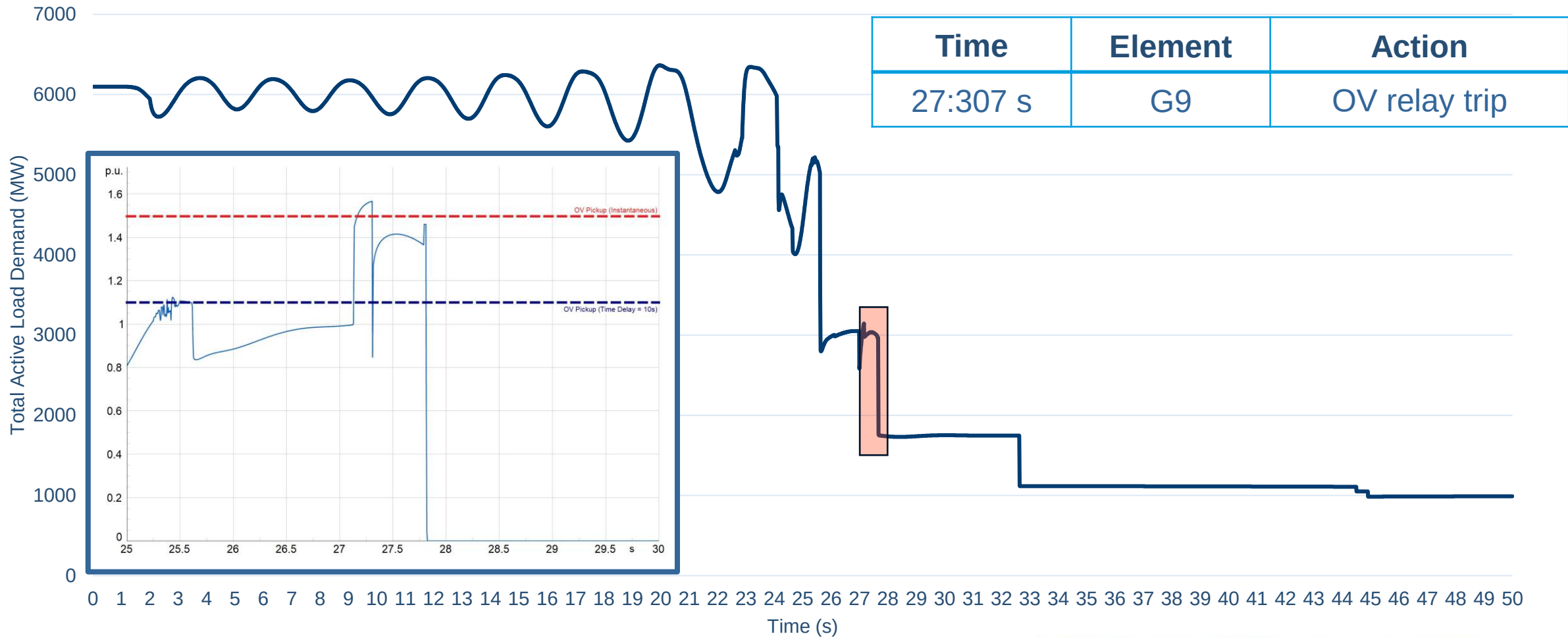
Cascading Failure Analysis: Attack on Substation (S2)

IEEE 39 Power Grid Loads



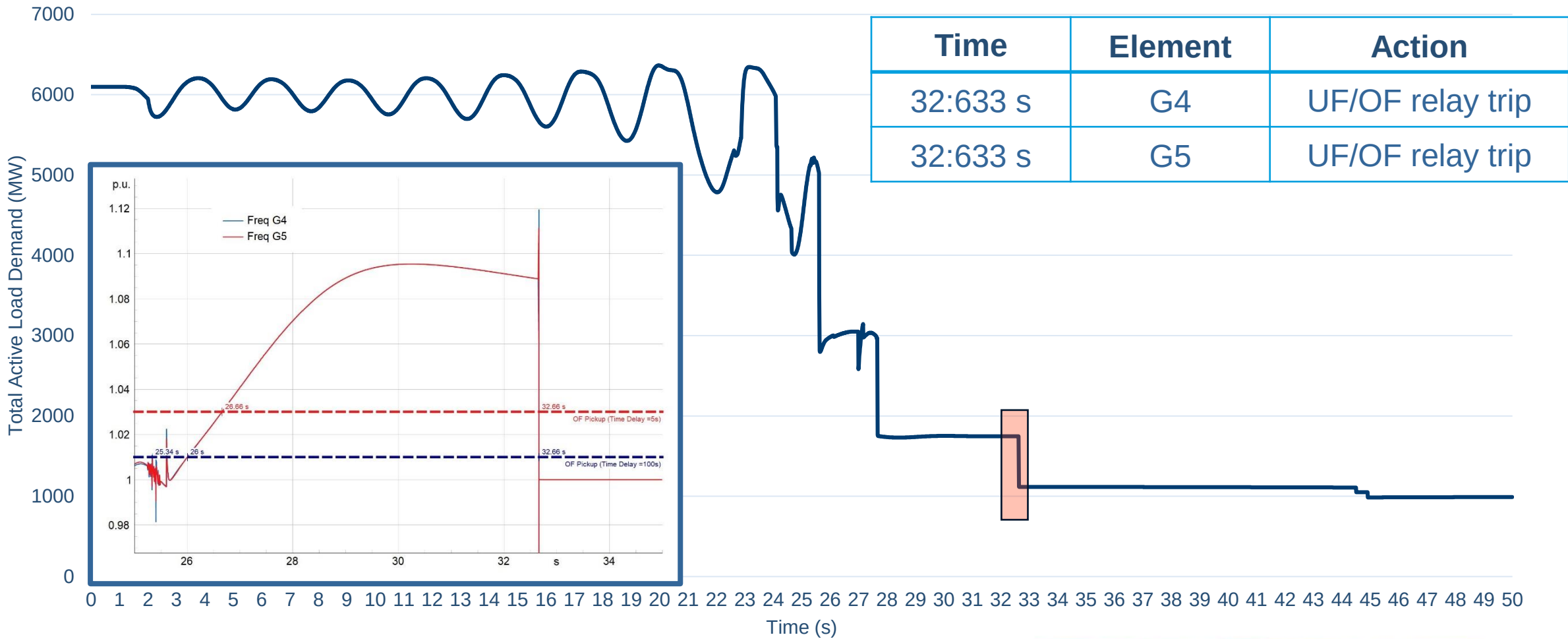
Cascading Failure Analysis: Attack on Substation (S2)

IEEE 39 Power Grid Loads

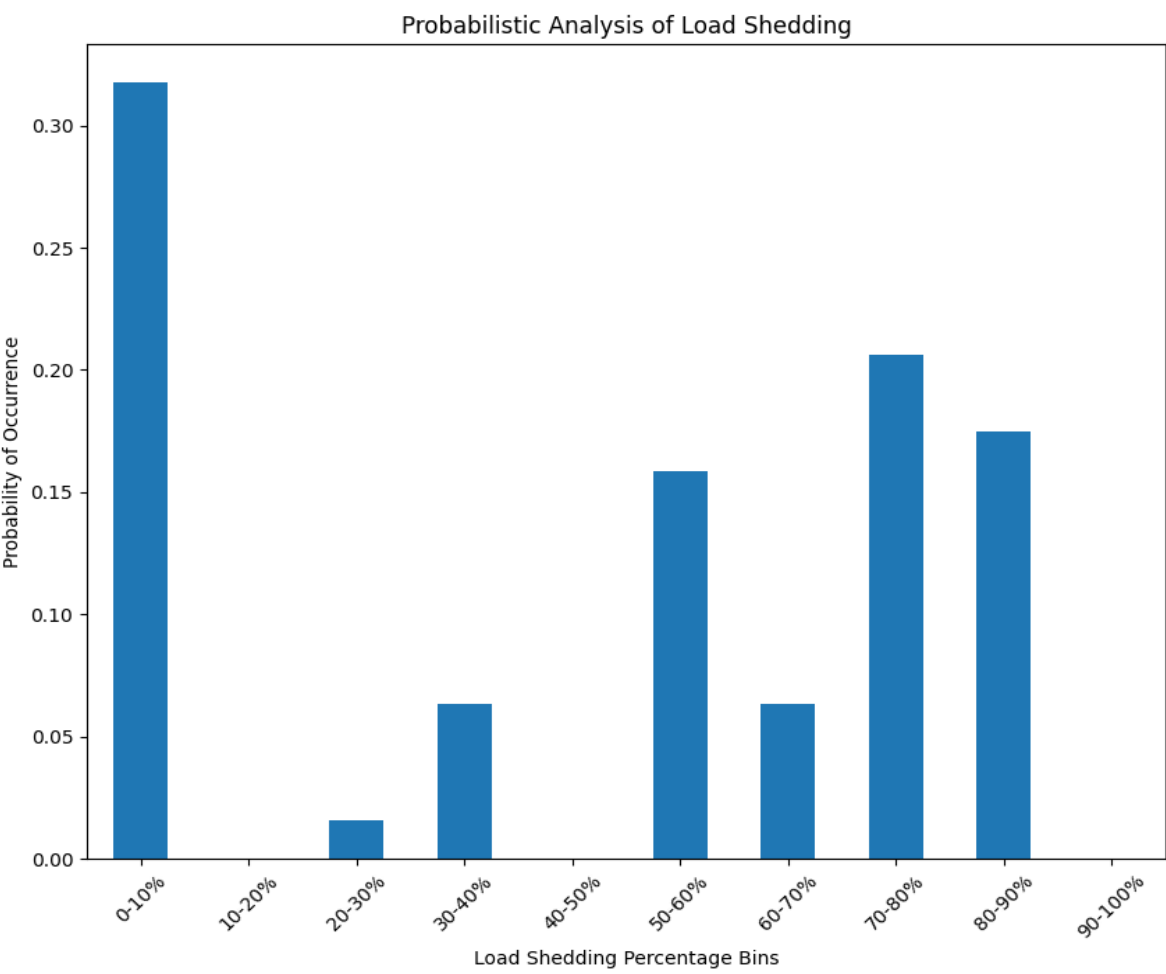
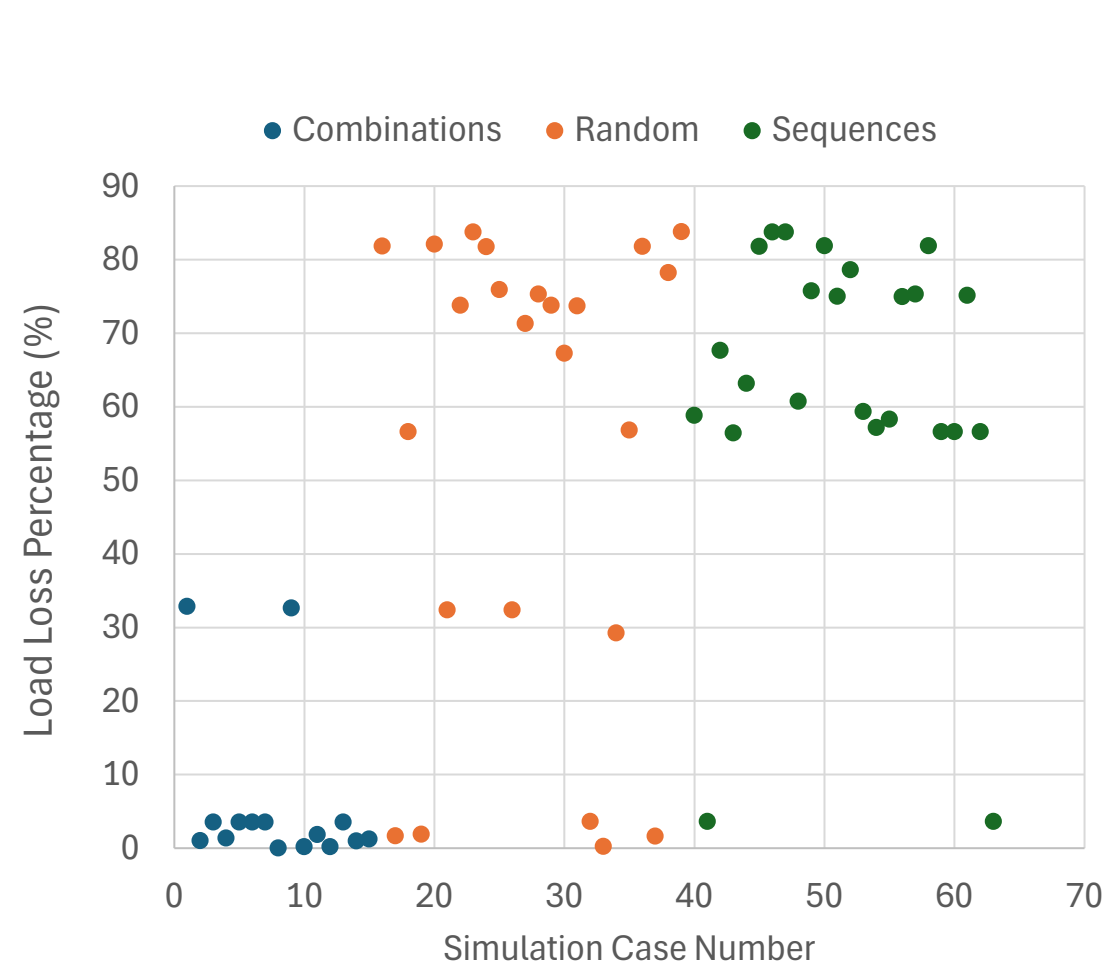


Cascading Failure Analysis: Attack on Substation (S2)

IEEE 39 Power Grid Loads



Cascading Failure Analysis: Attack on Substation (S2)



Cascading Failure Analysis: Cyber Attack Impact Index

Cyber attack impact index

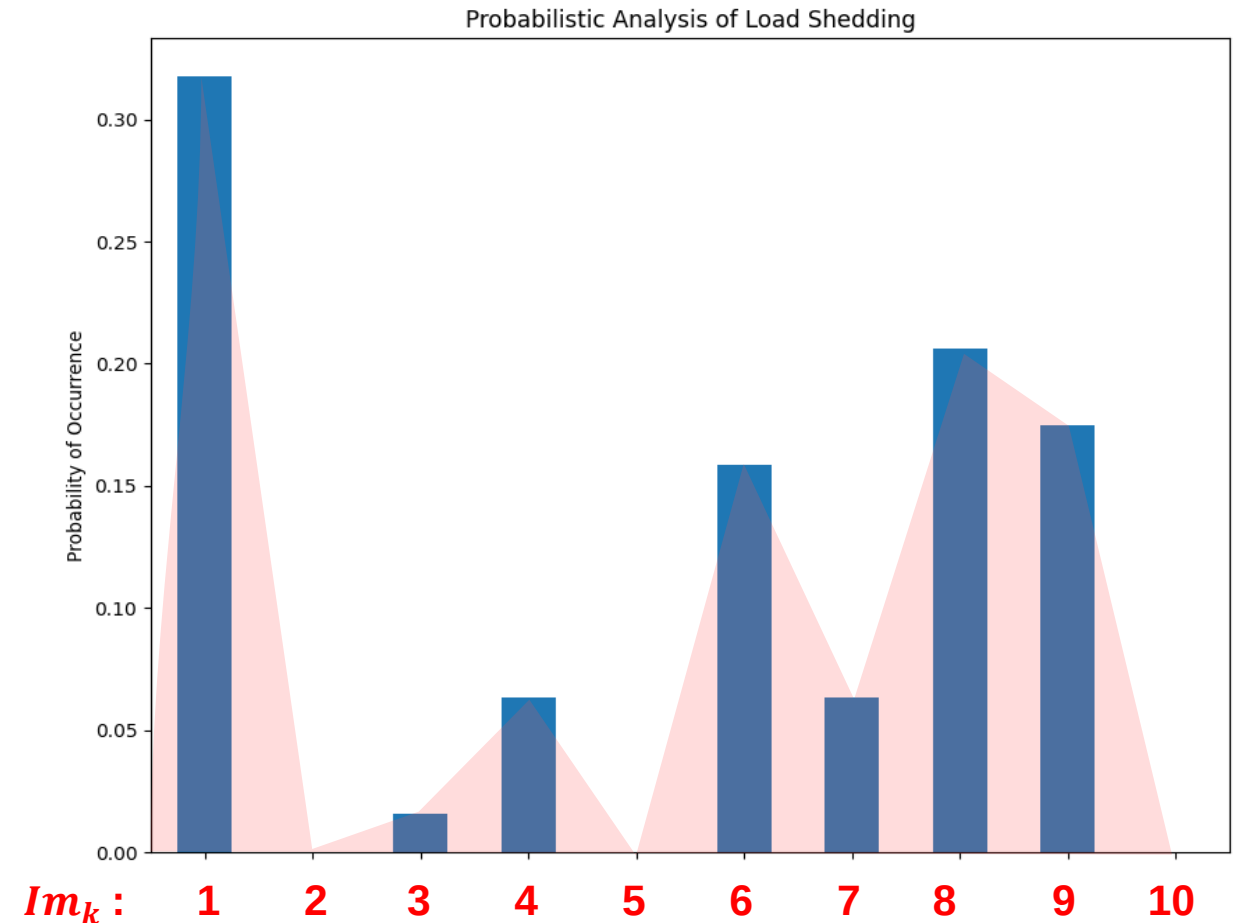
1. Assign weights to the impact of each bin (Im_k).
2. Calculate the probability of each bin using:

$$P_k = \frac{\text{Number of occurrences of load loss in bin } (k)}{\text{Total number of simulations}}$$

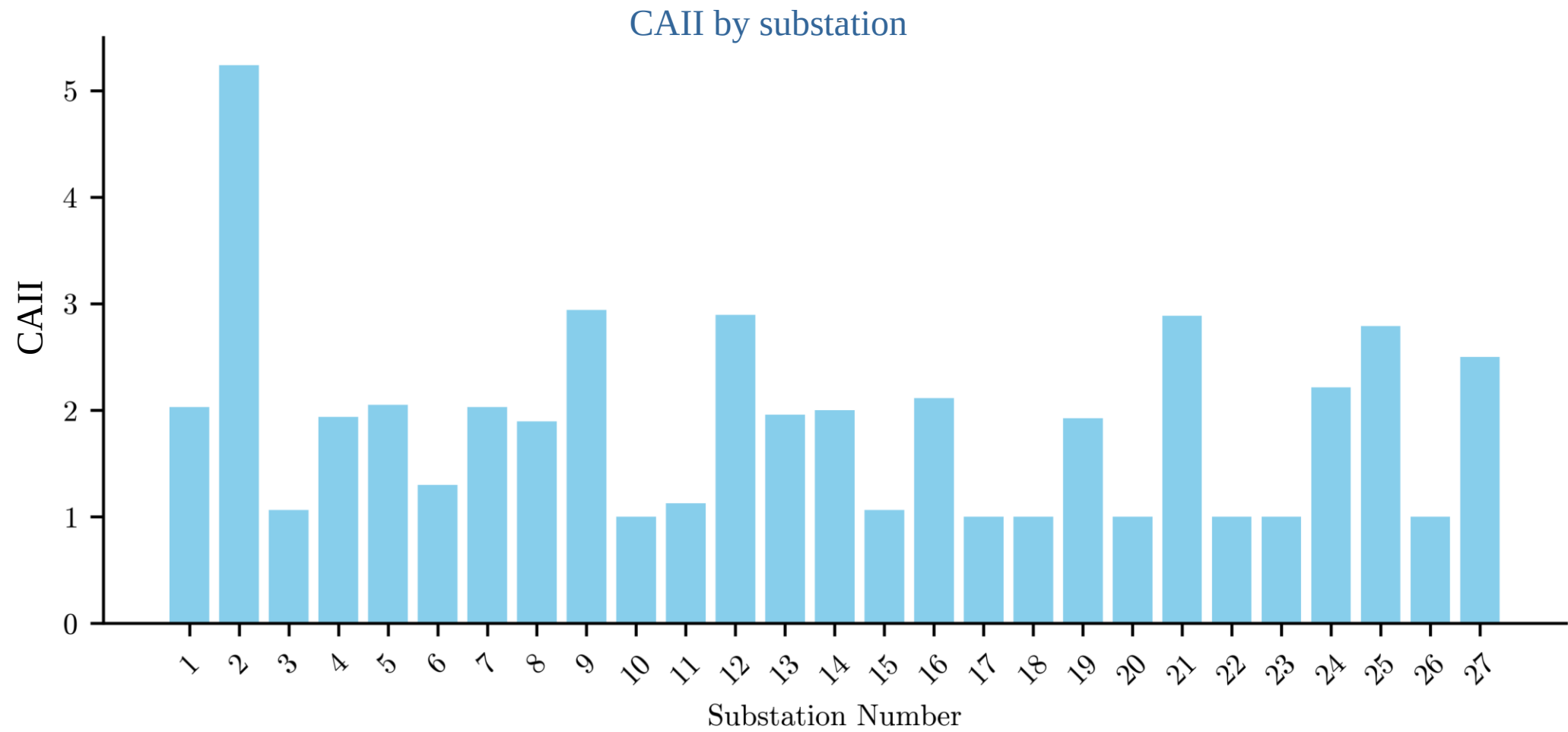
3. For each bin, multiply the probability (as a fraction) by its assigned weight. The Risk Index is then given by:

$$CAII = \sum_{k=1}^{10} P_k * Im_k$$

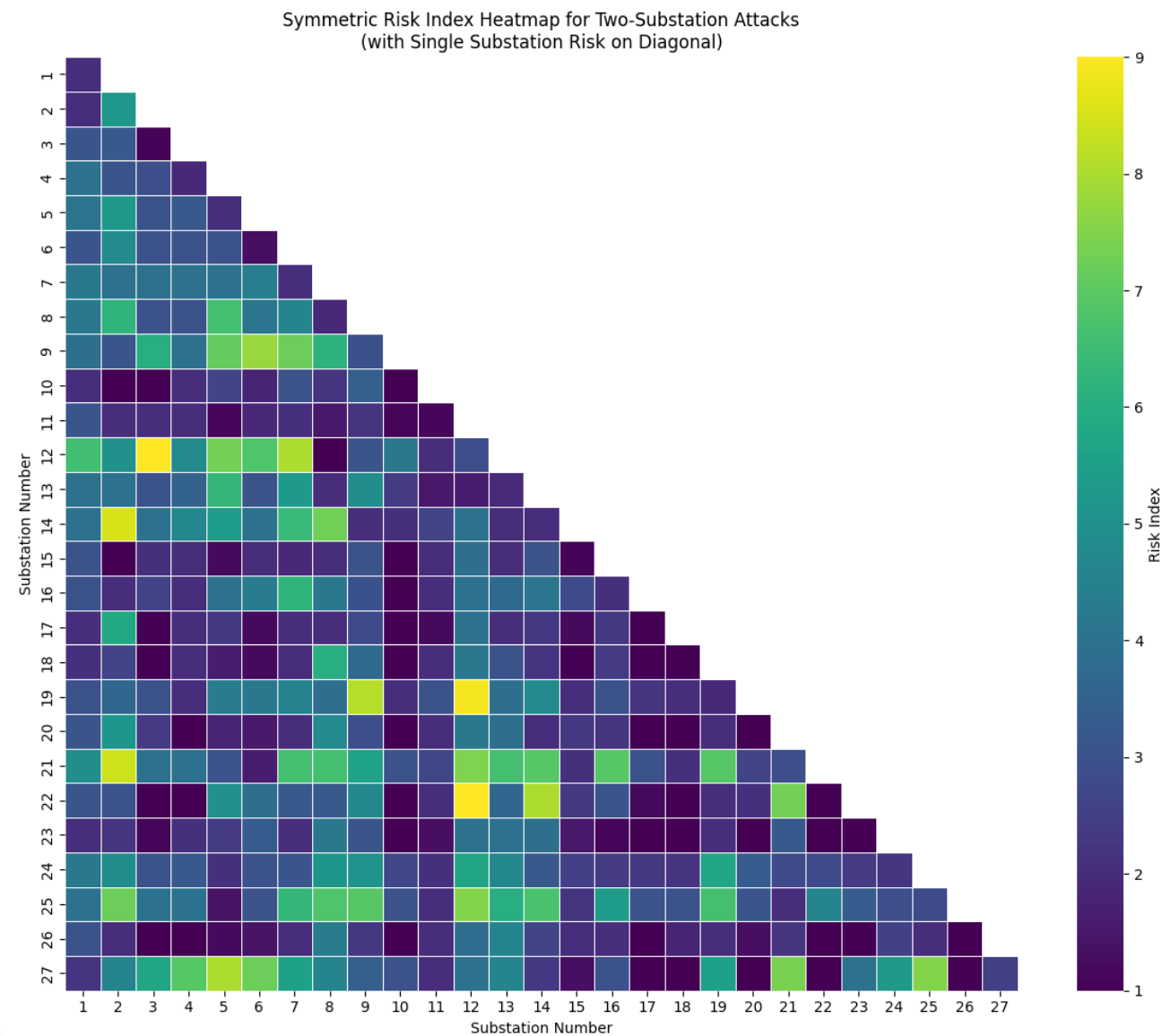
For Substation 2: CAII = 5.24



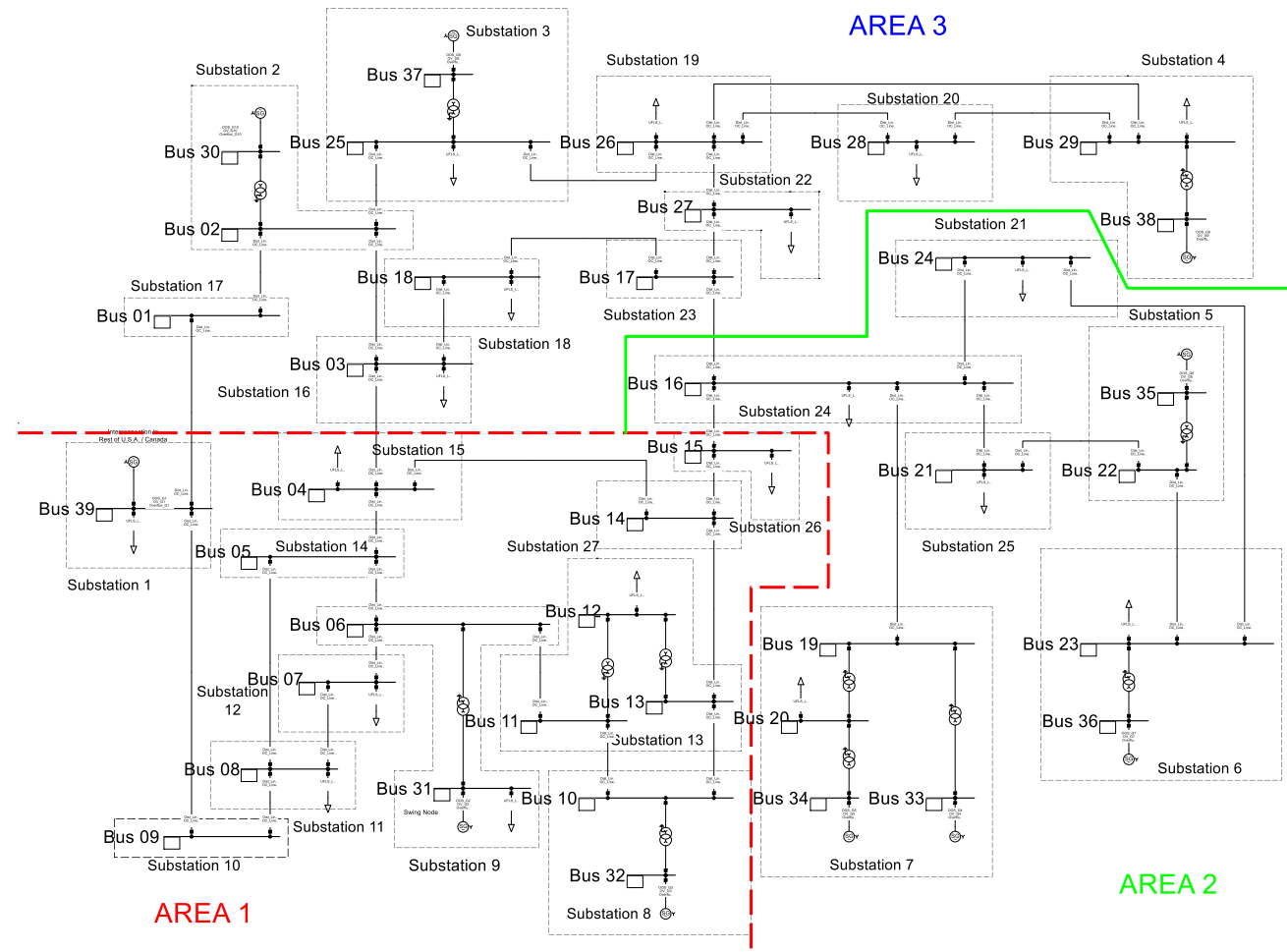
Cascading Failure Analysis: Cyber Attack Impact Index



Cascading Failure Analysis: Cyber Attack Impact Index



Cyber-Physical Attack

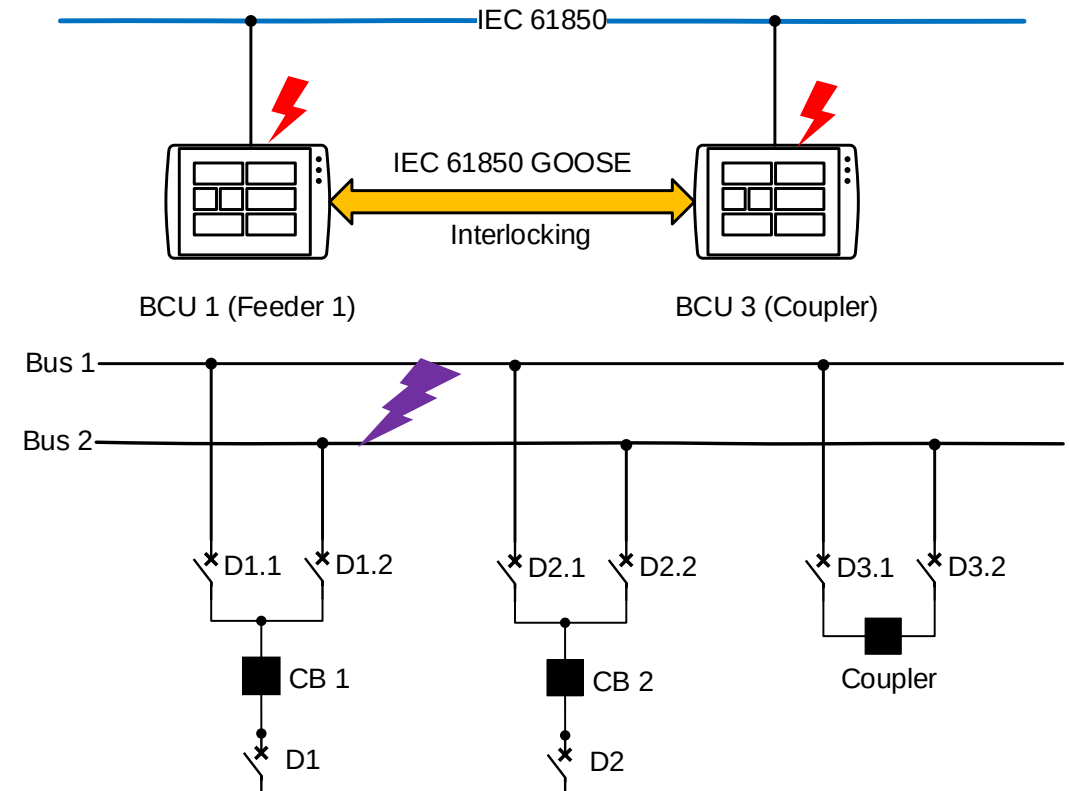


Digital Substation Interlocking Spoofing

- Coupler and circuit breaker statuses sent to the Bay Control Unit (BCU)
- Prevent inadvertent opening of disconnects under load
- GOOSE spoofing → open disconnects under load
- Bolted busbar fault and electric arc in the substation

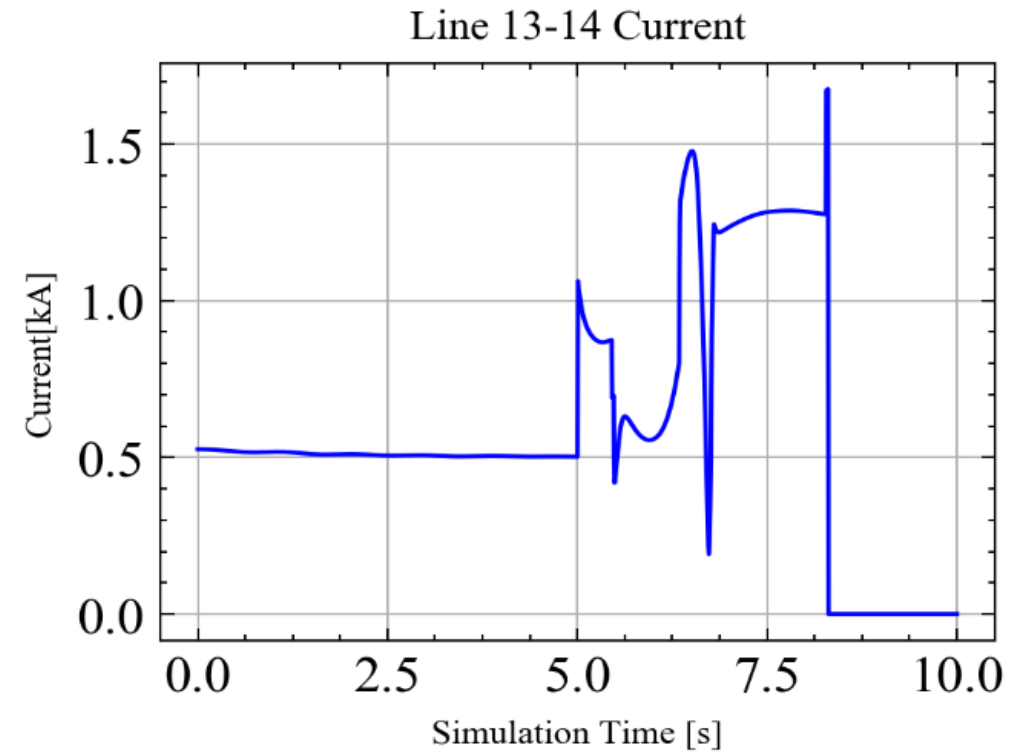
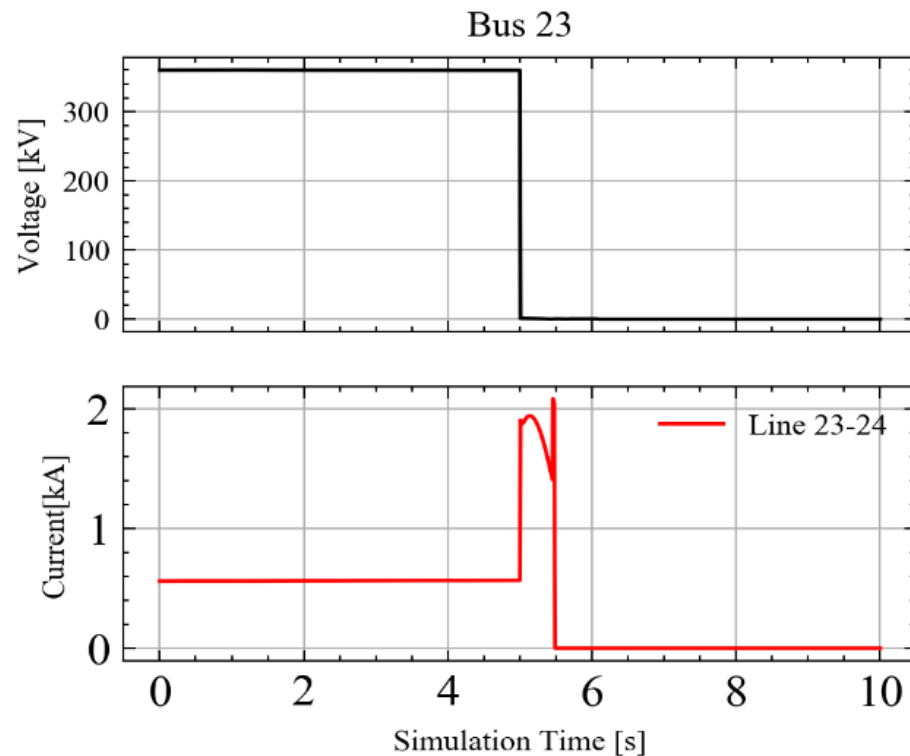


150kV HV station in Dronten, NL; 2022-09-02



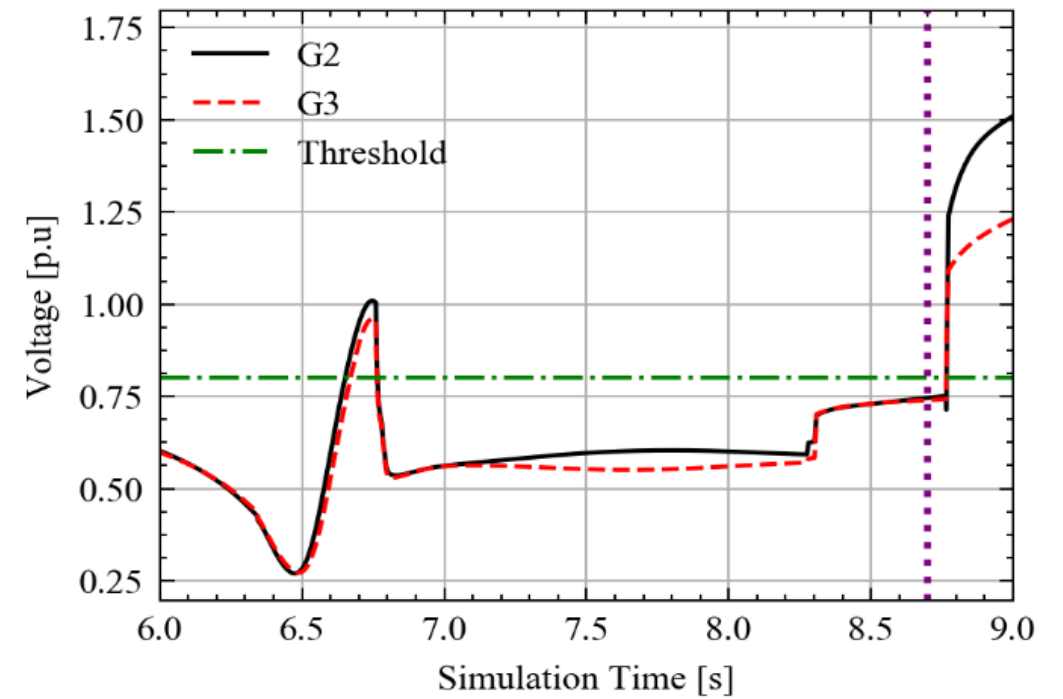
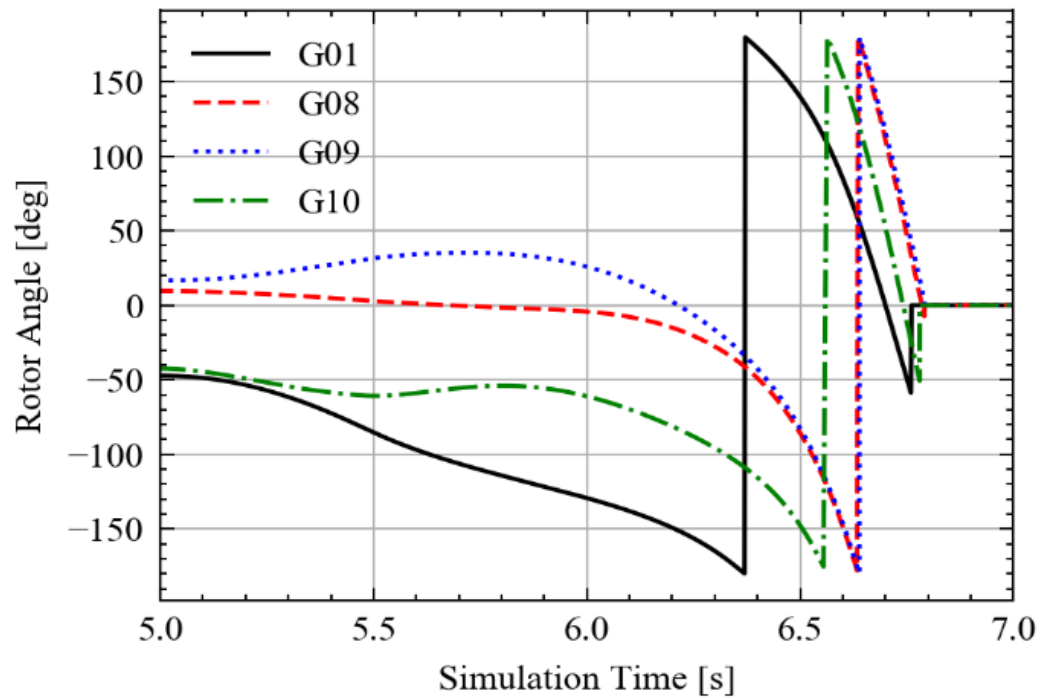
Cascading Failures: Cyber-Physical Attack

- Interlocking spoofing: busbar fault on bus 23 at 5s simulation time
- Protection DoS: increasing fault clearing time



Cascading Failures: Cyber-Physical Attack

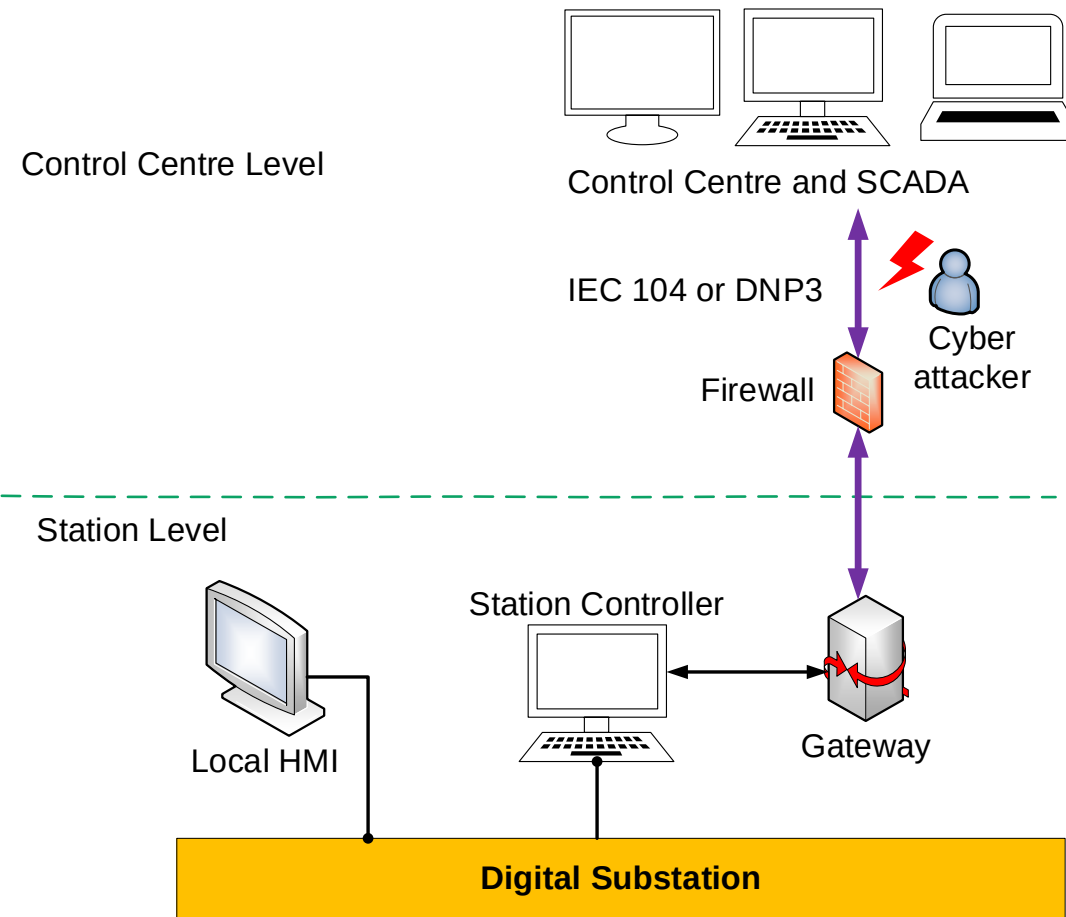
- Loss of synchronism of multiple generators
- Under voltage trip of G2 and G3 at 8.7s simulation time



Cascading Failures: Sequence of Events

Time	Event
5s	Cyber attack causes busbar fault on bus 23.
5.5s	Multiple lines in vicinity of attack location tripped by distance protection, i.e., lines 21-22 and 23-24.
6.1s	Generator G6 is islanded and tripped by ROCOF protection.
6.1 – 6.3s	Multiple lines tripped by distance protection, i.e., lines 22-23 and 16-19. PNR is reached.
6.6 – 6.7s	Lines 13-14 and 04-05 tripped by distance protection.
6.8s	Multiple generators, i.e., G1, G8, and G9 lose synchronism.
7.3s	Generator G10 is islanded and tripped by ROCOF protection.
8.7s	Generators G3 and G2 are disconnected by under voltage protection.
10s	End of simulation. Cyber attack results in loss of load ~ 6000 MW.

Hypothetical Hijacking Attack Scenario



No.	Event
1.	Cyber attack at three substations. Transformer tap positions are tampered causing changes to voltages.
2.	Generators automatically increase Var outputs. System operator is sent spoofed packets from compromised substations.
3.	Automatic UVLS is activated to alleviate low voltages.
4.	Attackers open multiple feeder circuit breakers at once, disconnecting numerous lines and loads.
5.	Sudden disconnections cause power swings and frequency instabilities. Generators are tripped by automatic overexcitation protection due to voltage instabilities.
6.	UFLS is activated to arrest frequency drop due to loss of generation.
7.	Three parallel overhead lines are overloaded beyond thermal limits and trip.
8.	Severe frequency and voltage instabilities cause cascading failures and blackout.

Summary and Takeaways

Cyber attacks can significantly impact power grid

- Loss of load
- Equipment damage: excessive torques physically damage generators
- Voltage and rotor angle instability
- Cascading outages/failures
- Partial or a complete blackout

Power system blackouts

- Europe & USA **112 million people** (2003)
- Europe **15 million people** (Nov. 2006)
- USA **2.7 million people** (Sep. 2011)
- India **600 million people** (Jul. 2012)
- Ukraine **225,000 customers** (Dec. 2015)



Thank You



Alex Stefanov

Email: A.I.Stefanov@tudelft.nl

Assistant Professor, Chartered Engineer (CEng MIEI)

Intelligent Electrical Power Grids, EEMCS, **TU Delft**
Mekelweg 4, 2628 CD Delft, Netherlands